

Critical Design Review

Leah, Megan, Athena, David, & Steven

Get Hype!

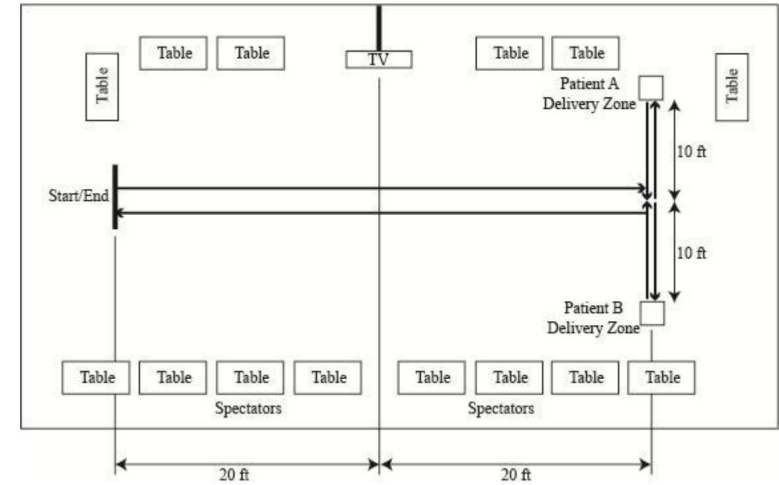


Agenda

- Project Overview
- Solution
- Build Process
- CAD
- Subsystem Breakdown
 - Chassis
 - Drivetrain
 - Electronics
 - Pill Dispenser
 - Cup Interface
 - Cover
 - Wearable Devices
 - Koozie
- Budget
- Reflection

Project Goals

1. Biomechanical Interaction: Correctly interpret a set biomechanical movement of Patient A's upper extremity and send a signal to the delivery robot
2. Thermal Interaction: Correctly interpret Patient B exceeding a set thermal threshold and send a signal to the delivery robot
3. System Agility/Speed: Successfully navigate the course based on the received input without hitting any tables or deviating from the designated paths
4. Medication Delivery: Move a set number of "pills" to the correct patient delivery zone and autonomously deliver the correct medication based on the received input

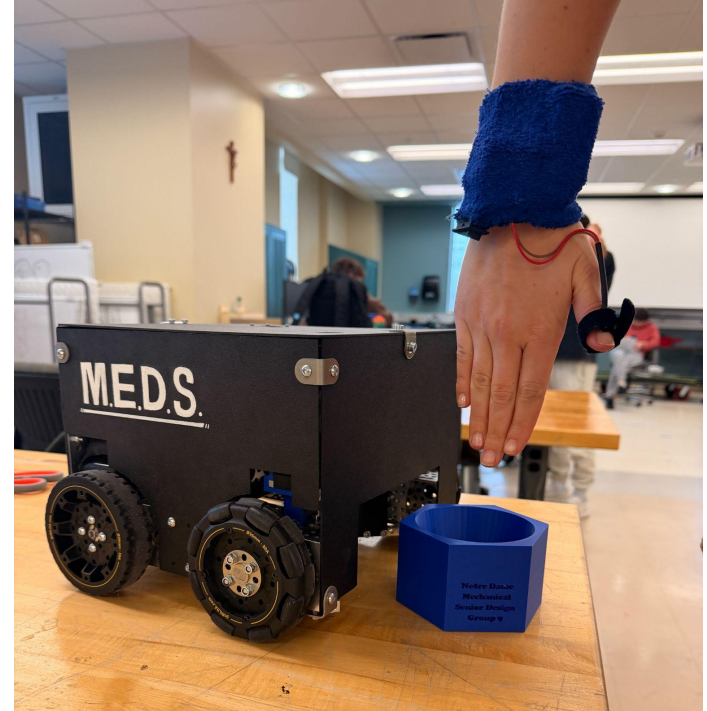


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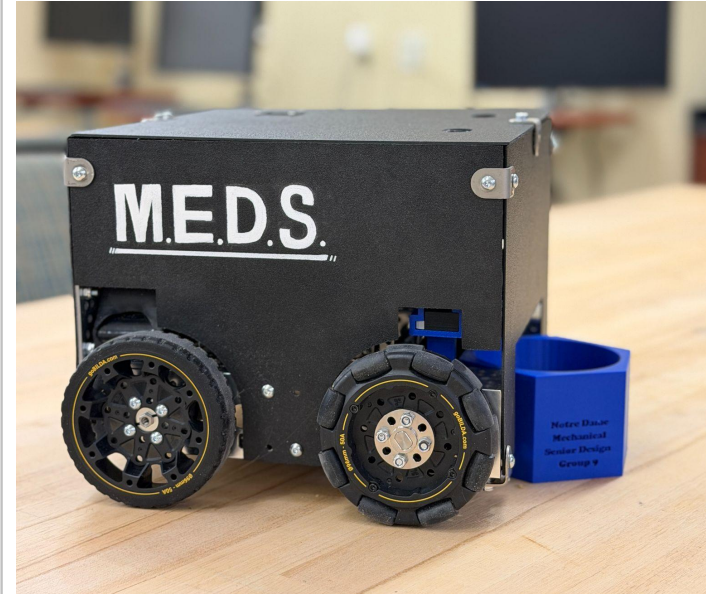
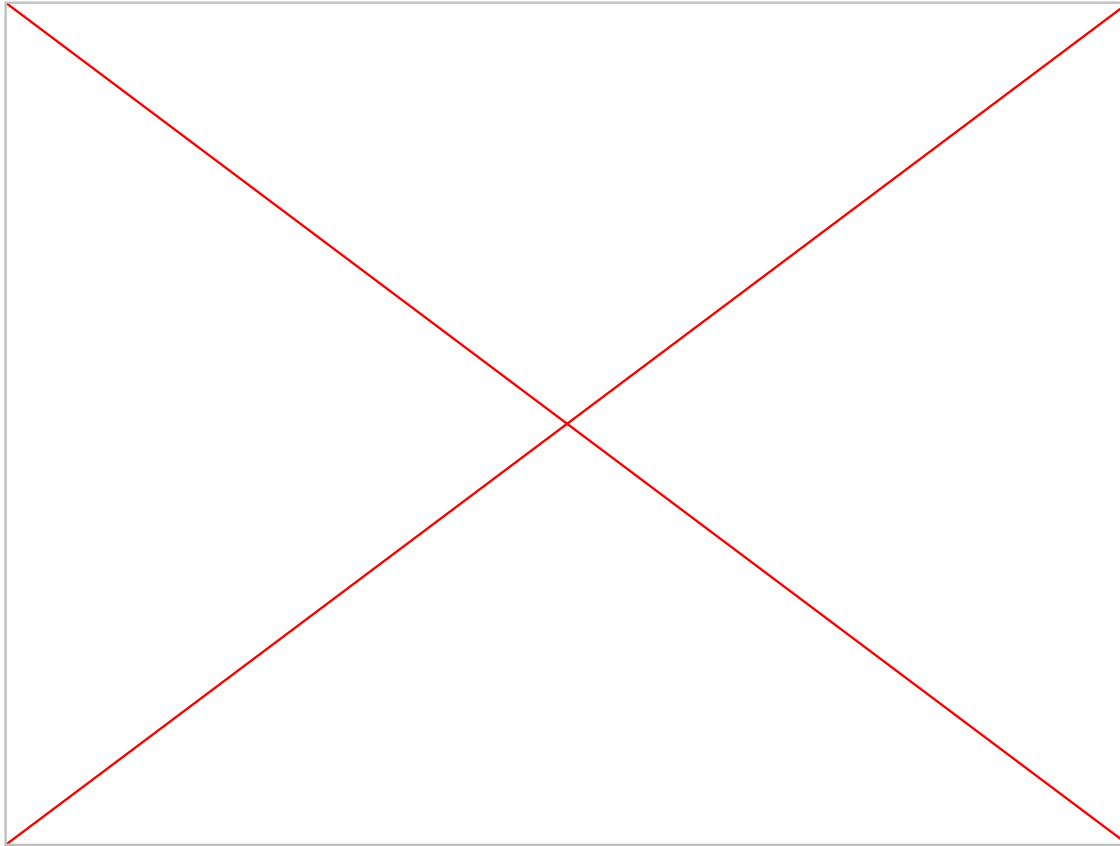
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Solution

1. This team created a Medical Express Delivery System (MEDS) that consists of three main subsystems: the Delivery Robot, the Wearable Device, and the Koozie.
2. The Delivery Robot navigates the course and delivers the medication.
3. The Wearable Device transmits a signal to the Delivery Robot when medication is needed.
4. The Koozie serves as a docking station between the patients' cups and the Delivery Robot



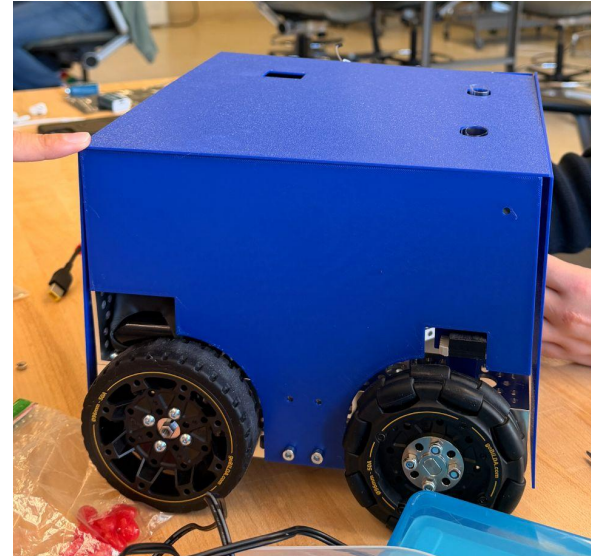
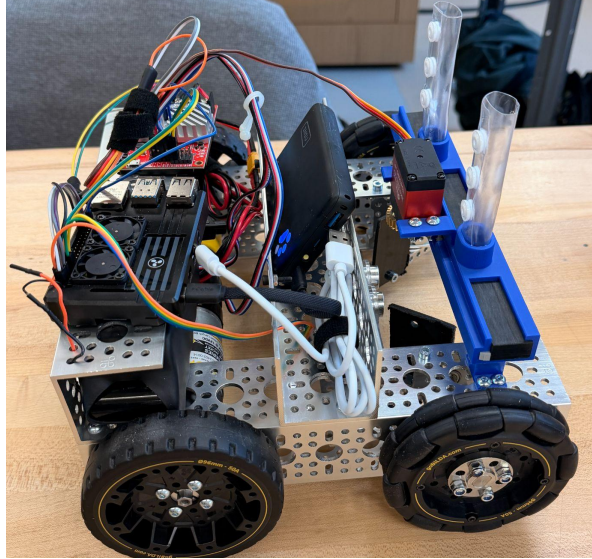
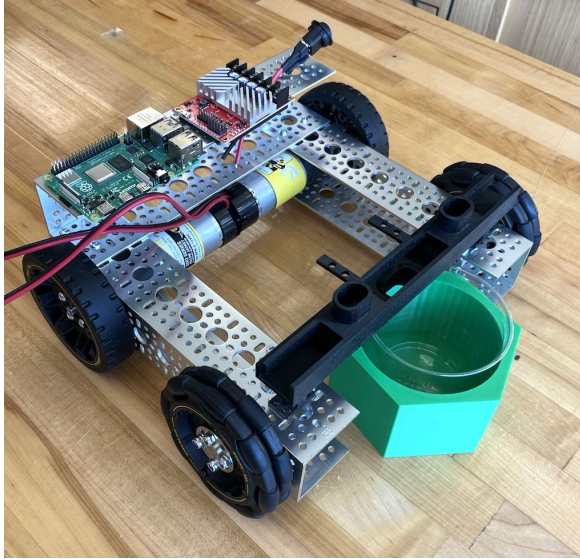
Final Robot System



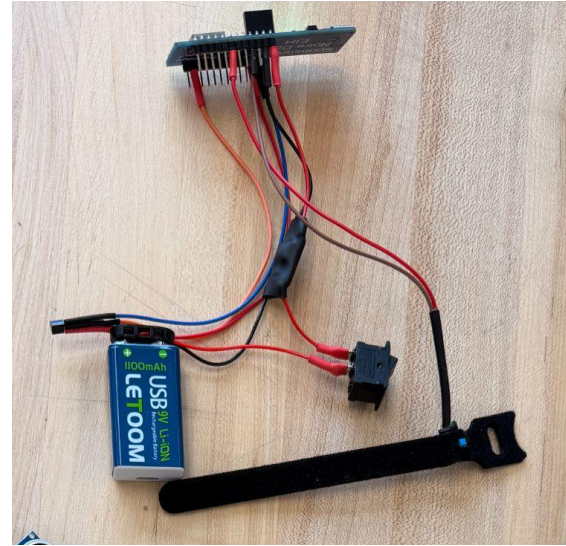
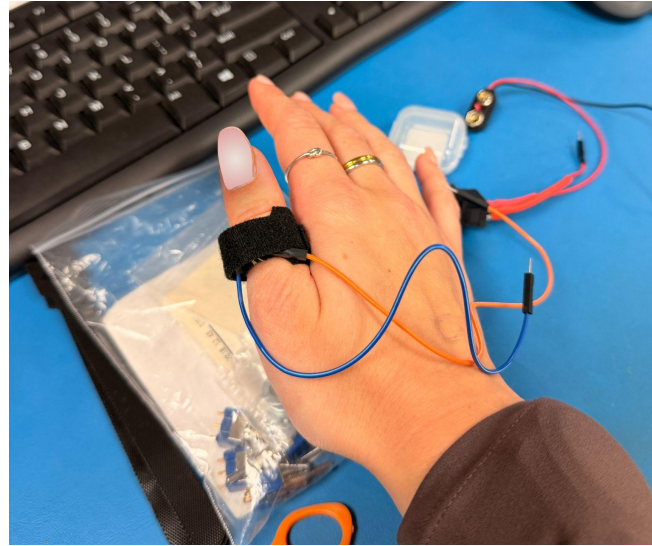
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Build Process - Delivery Robot



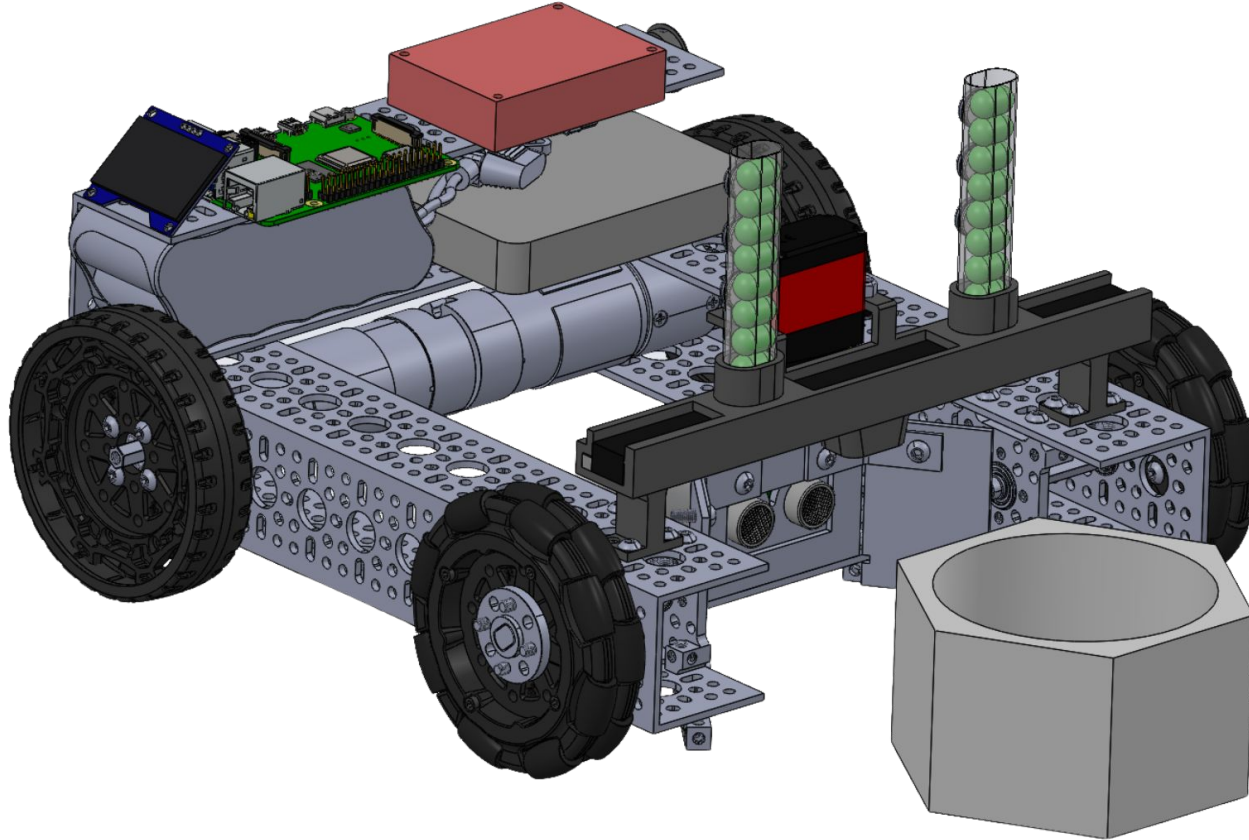
Build Process - Wearable Device



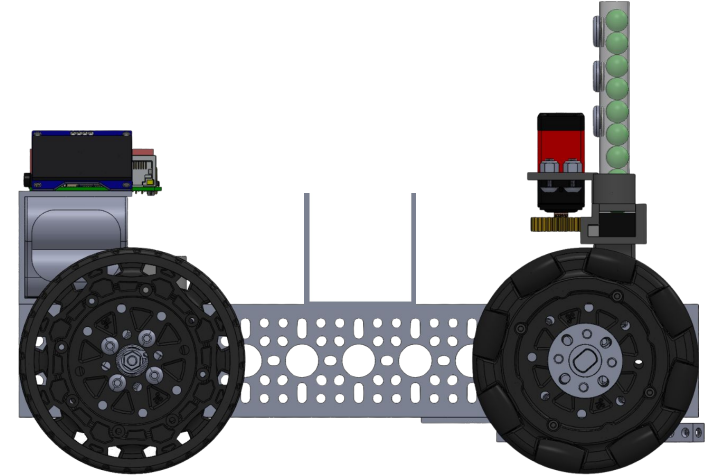
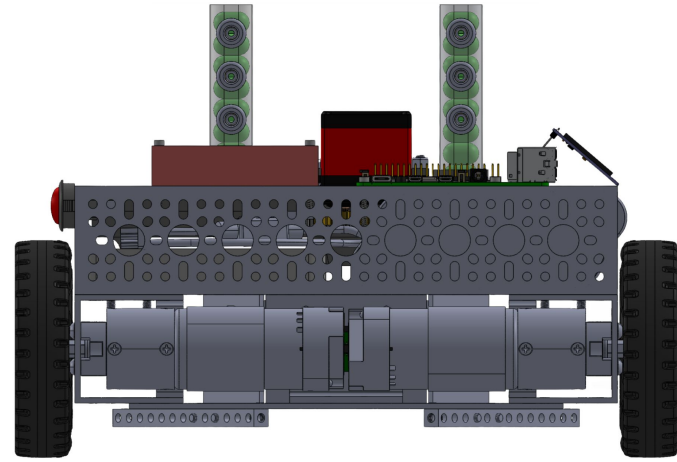
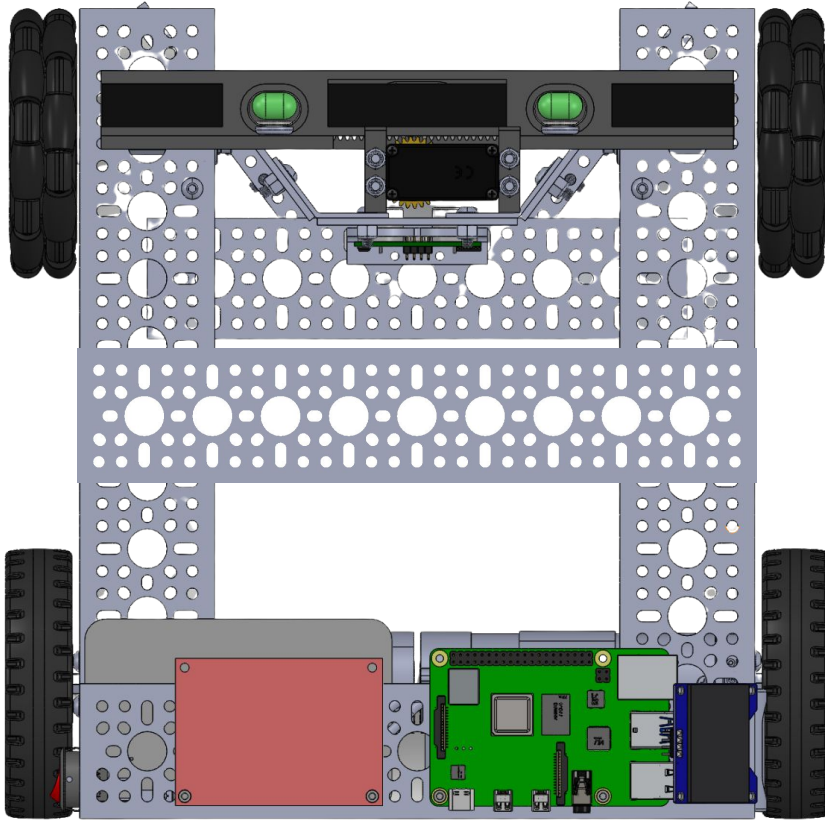
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Completed Virtual Model - Internal Subsystems



Completed Virtual Model - Top/Rear/Side

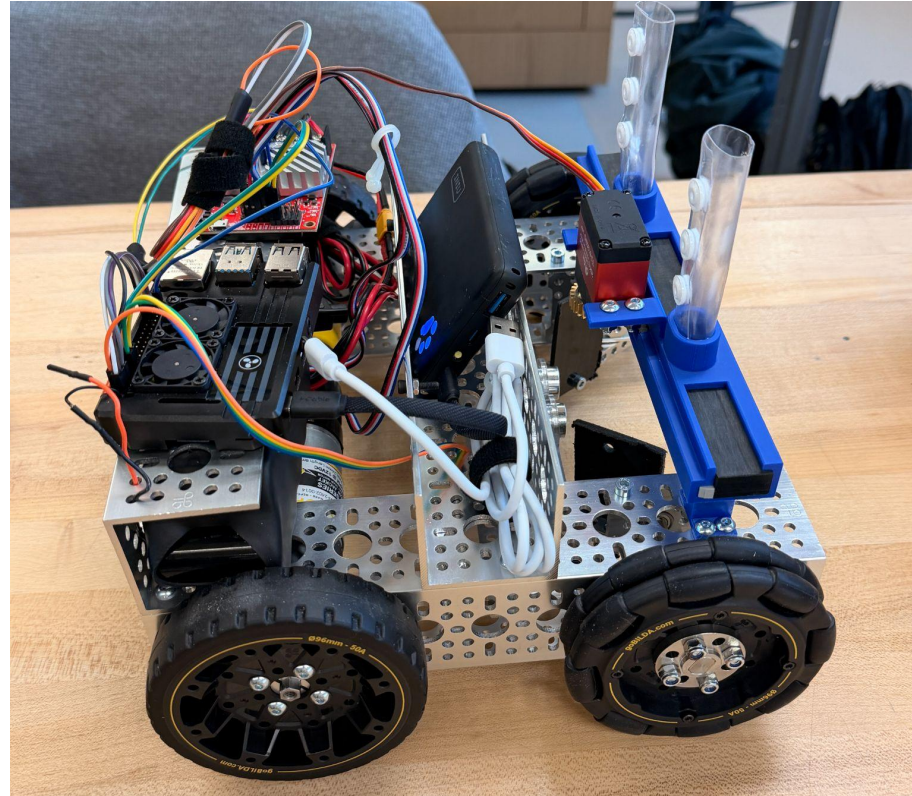


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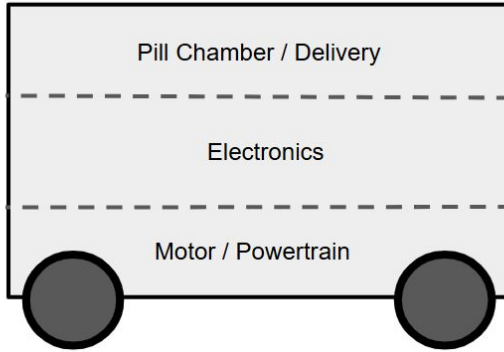
Chassis - Final Design

Final construction of
the chassis

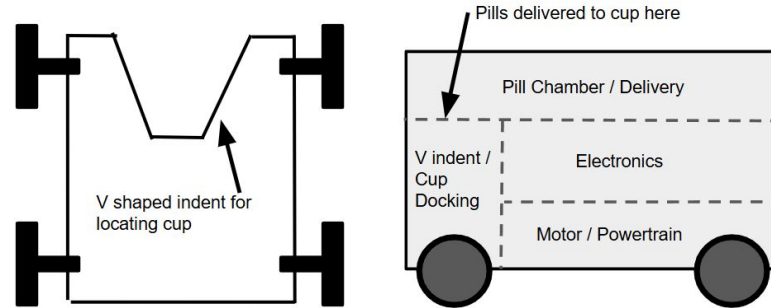


Chassis - Brainstorming

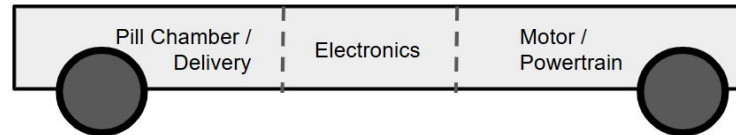
Concept 1 - Layered Chassis



Concept 2 - V Frame Chassis



Concept 3 -Low Profile Chassis



Chassis - Critical Design Decisions

Chassis Design and Dimensions

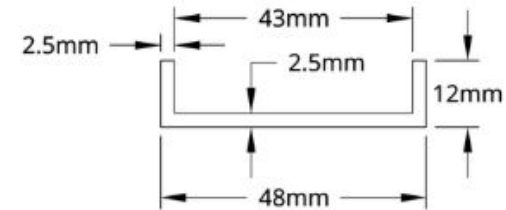
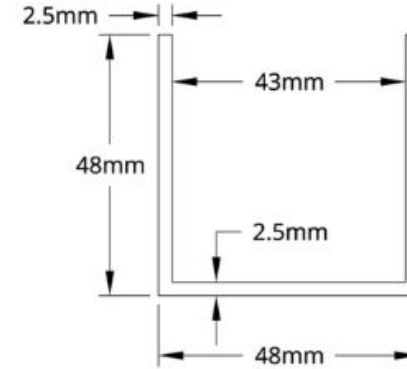
- Square by request of electronics lead
- Must fit within 12"x12"
- Minimum of 9.25" because of motors, less than 10" because of wheels

Beam Profile

- Must be able to interface with and support the other subassemblies
- C-Channel vs U-Channel

Beam Material

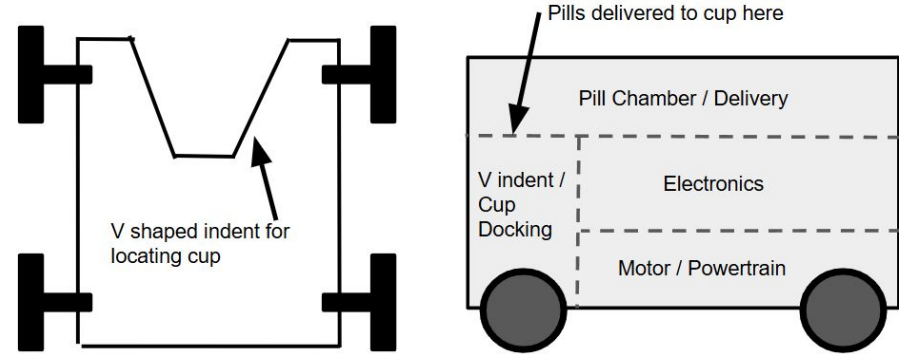
- 7005-T6 Aluminum vs 6063-T6 Aluminum



Chassis - Trade Study Results

- Design Variables:
 - Beam Profile
 - Chassis Material
- State Variables:
 - Yield Strength Factor of Safety (40%)
 - Deflection (30%)
 - Weight (15%)
 - Cost (15%)
- Model and Equations
 - MATLAB
 - Solid Mechanics, Design of Machine Elements, Statics
 - Deflection, von Mises Stress, Moment of Inertia, Shear Stress, Bending Stress

Concept Design



$$\delta = -\frac{5wl^4}{384EI}$$

$$\sigma' = \sqrt{\sigma^2 + 3\tau^2}$$

$$\tau = \frac{VQ}{It}$$

$$\sigma = \frac{My}{I}$$

$$I_{\text{channel}} = \frac{wh^3 - (w-t)(h-3t)^3}{12}$$

Chassis - Trade Study Decision Matrix

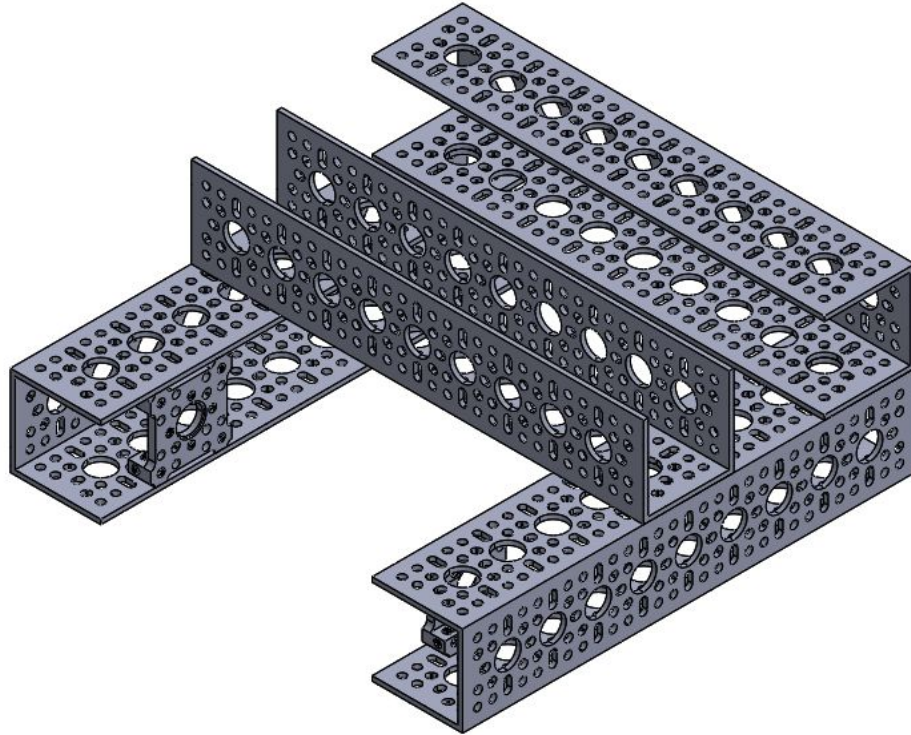
Determine the alternative that best meets the engineering specifications	State Variable Results (1-10 Scale)								
	Yield Strength Factor of Safety		Deflection		Weight		Cost (one beam)		Total
	0.4		0.3		0.15		0.15		1
Designs	Absolute	Weighted	Absolute	Weighted	Absolute	Weighted	Absolute	Weighted	Weighted
Beam 1 (C-Channel, 7005 T6)	3.43	1.37	2.53	0.76	5.74	0.86	4.67	0.70	3.69
Beam 2 (U-Channel, 7005 T6)	8.37	3.35	9.68	2.90	1.72	0.26	4.01	0.60	7.11
Beam 3 (C-Channel, 7005 T6)	4.27	1.71	4.04	1.21	5.02	0.75	7.33	1.10	4.77
Beam 4 (U-Channel, 7005 T6)	7.70	3.08	8.86	2.66	1.80	0.27	6.33	0.95	6.96
Beam 1 (C-Channel, 6063 T6)	1.73	0.69	2.42	0.73	6.04	0.91	4.67	0.70	3.03
Beam 2 (U-Channel, 6063 T6)	5.03	2.01	9.64	2.89	2.28	0.34	4.01	0.60	5.85
Beam 3 (C-Channel, 6063 T6)	2.30	0.92	3.95	1.18	5.36	0.80	7.33	1.10	4.01
Beam 4 (U-Channel, 6063 T6)	4.57	1.83	8.81	2.64	2.36	0.35	6.33	0.95	5.77

- Concept Recommendation
 - goBilda U-Channel beam (4x) - profile handles bending and von Mises stress the best of all beams
 - 7005-T6 Aluminum - high strength to density ratio at a moderate cost and lower weight

Chassis - Design Changes

1. Center beam lengthened and placed on top of the side beams instead of directly in the center. This increased connection simplicity and gave more functional space.
2. Added a U-channel in the center to secure the Raspberry Pi battery and IMU. This allowed for additional electronics storage and structural stability.
3. Added hardware and brackets to help mount electronics and sensors
4. Mounted a box cover to protect components and enhance aesthetics.

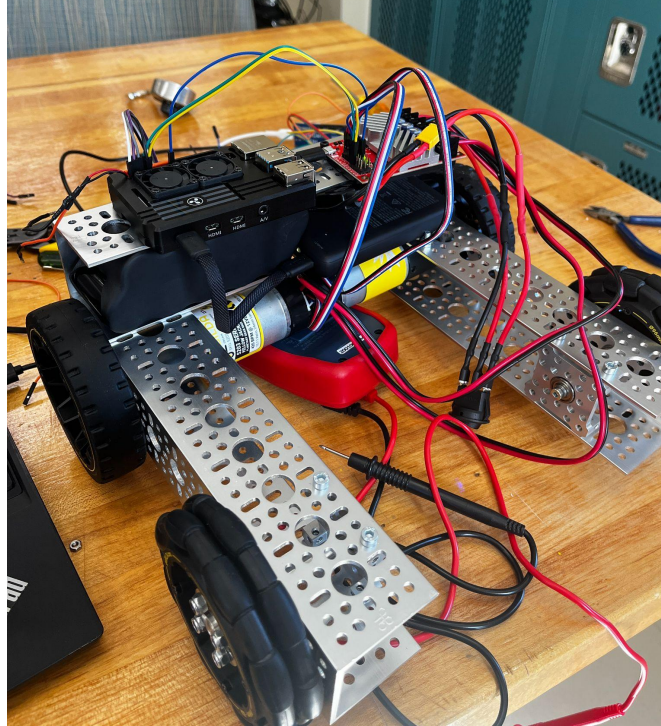
Chassis - Final CAD



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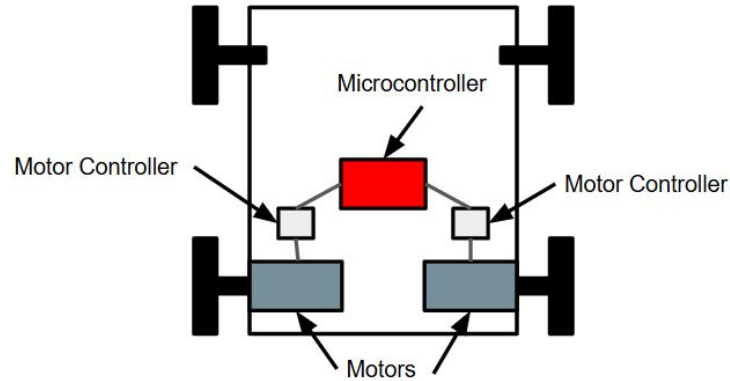
Drivetrain - Final Design



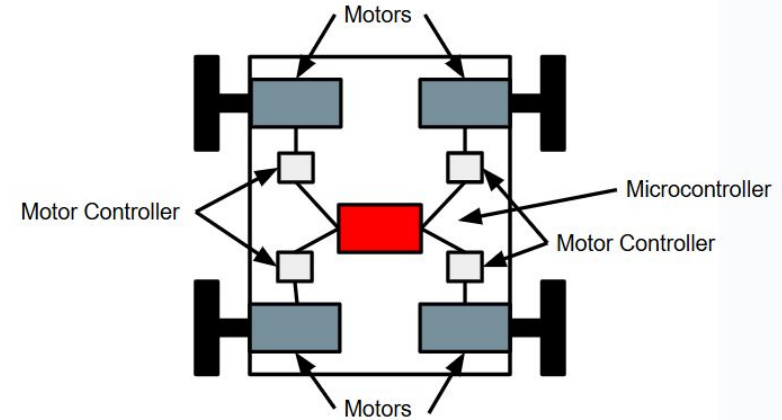
Final construction of the drivetrain

Drivetrain - Brainstorming

Skid Steering



Mecanum Drive



Drivetrain - Critical Design Decisions

Driving Mechanism

- Skid Steering was chosen for simplicity and to stay within budget. It only requires 2 motors instead of 4 and only 1 motor controller which greatly decreased cost and complexity

Motor Selection

- Delivery device must maximize velocity without exceeding 5 mph
- Motors must be able to support up to 50 lbs

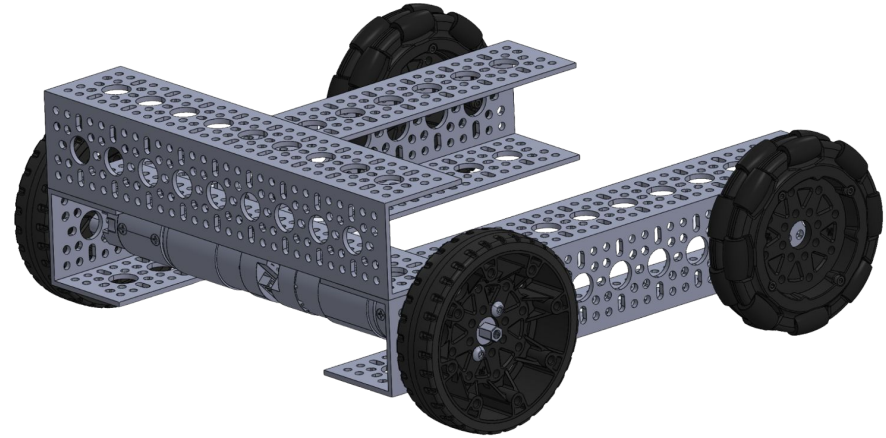
Wheels

- Interface with chassis and motors
- With chassis must fit 12"x12"
- Back and front wheels must have same diameter

Drivetrain - Trade Study Results

- Design Variables:
 - Motor
 - Wheels
- State Variables:
 - Time to Complete Course (70%)
 - Propulsion Force Factor of Safety (15%)
 - Total Weight (10%)
 - Total Cost (5%)
- Model
 - MATLAB Simulation + SolidWorks Geometry Check
 - Kinematics and Acceleration Profiles
 - Torque vs. Speed
 - Friction and Wheel Slip

Concept Design



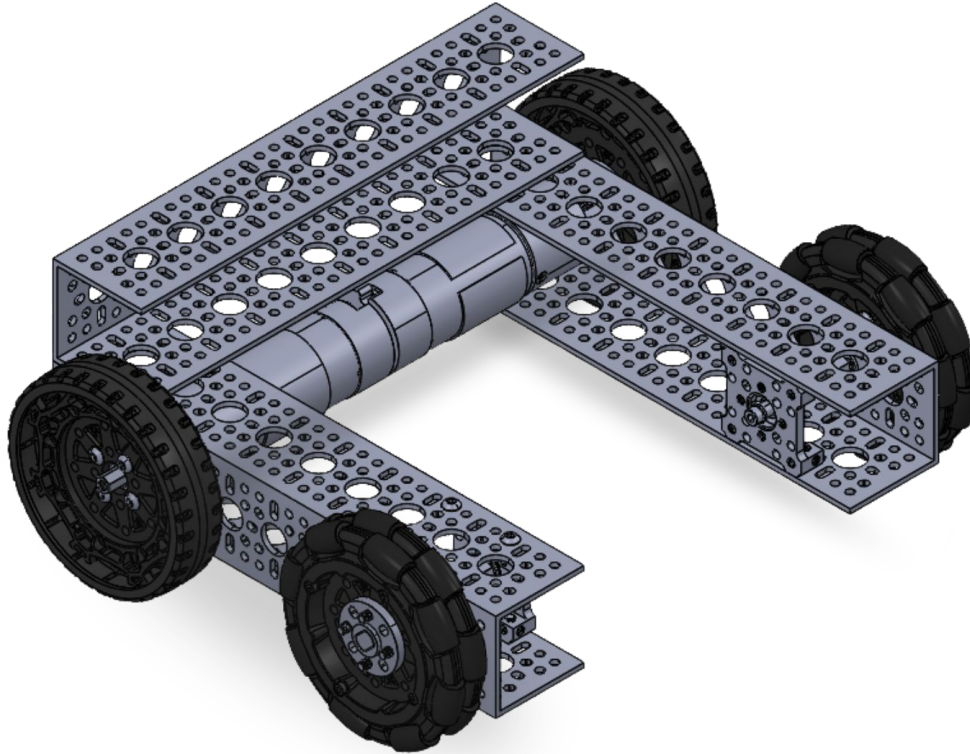
Drivetrain - Trade Study Decision

Determine the alternative that best meets the engineering specifications	State Variable Results (1-10 Scale)								
	Course Completion Time		Propulsive Force Factor of Safety		Weight of Components		Total Cost		Total
	0.7		0.15		0.1		0.05		1
	Absolute	Weighted	Absolute	Weighted	Absolute	Weighted	Absolute	Weighted	Weighted
Motor #1 with Wheel #1	9.98	6.98	0.00	0.00	3.48	0.35	2.32	0.12	7.45
Motor #1 with Wheel #2	8.83	6.18	0.00	0.00	4.49	0.45	2.68	0.13	6.77
Motor #1 with Wheel #3	9.99	6.99	5.00	0.75	0.00	0.00	0.72	0.04	7.78
Motor #1 with Wheel #4	10.00	7.00	10.00	1.50	0.58	0.06	0.00	0.00	8.56
Motor #2 with Wheel #1	8.62	6.04	0.00	0.00	3.48	0.35	2.32	0.12	6.50
Motor #2 with Wheel #2	7.01	4.91	0.00	0.00	4.49	0.45	2.68	0.13	5.49
Motor #2 with Wheel #3	8.63	6.04	5.00	0.75	0.00	0.00	0.72	0.04	6.83
Motor #2 with Wheel #4	8.64	6.05	10.00	1.50	0.58	0.06	0.00	0.00	7.61
Motor #3 with Wheel #1	3.38	2.37	0.00	0.00	8.99	0.90	9.64	0.48	3.75
Motor #3 with Wheel #2	0.00	0.00	0.00	0.00	10.00	1.00	10.00	0.50	1.50
Motor #3 with Wheel #3	3.39	2.37	5.00	0.75	5.65	0.57	8.04	0.40	4.09
Motor #3 with Wheel #4	3.39	2.37	10.00	1.50	6.23	0.62	7.32	0.37	4.86
Motor #4 with Wheel #1	8.43	5.90	0.00	0.00	8.99	0.90	9.64	0.48	7.28
Motor #4 with Wheel #2	6.75	4.73	0.00	0.00	10.00	1.00	10.00	0.50	6.23
Motor #4 with Wheel #3	8.44	5.91	5.00	0.75	5.65	0.57	8.04	0.40	7.63
Motor #4 with Wheel #4	8.45	5.92	10.00	1.50	6.23	0.62	7.32	0.37	8.40

Concept Recommendation

- 435 RPM GoBuilda Motors (2x)
- 96mm Hogback Traction Wheels (2x, Driven)
- 96mm Omni Wheels (2x, Free-Spinning)
 - Omni wheel properties held constant except for diameter, which matched the driven wheels

Drivetrain - Final CAD



The drive train was not altered
after the CAD was finalized

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Electronics Diagram

x2 12V motors with encoders and RoboClaw motor controller

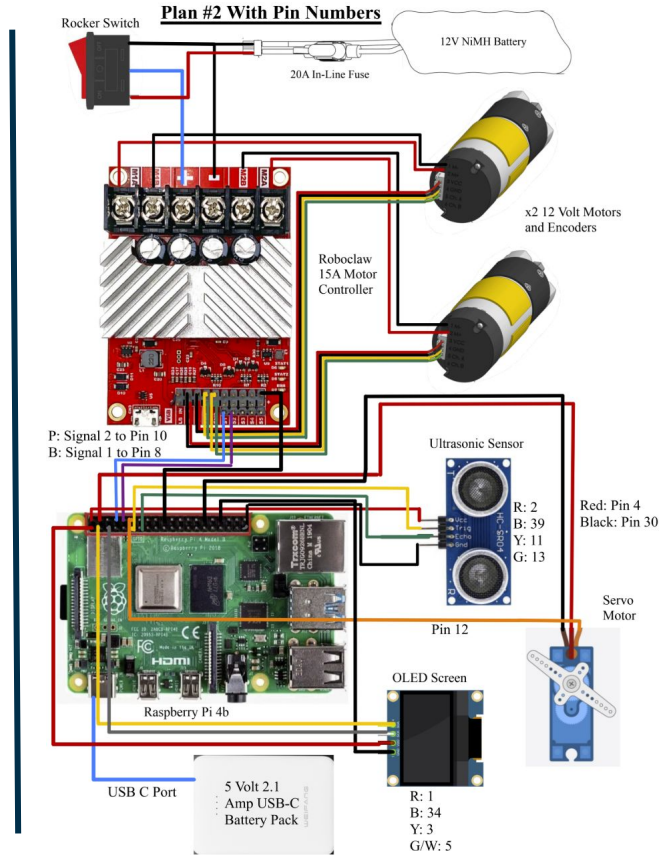
Microcontroller:

- Raspberry Pi 4b

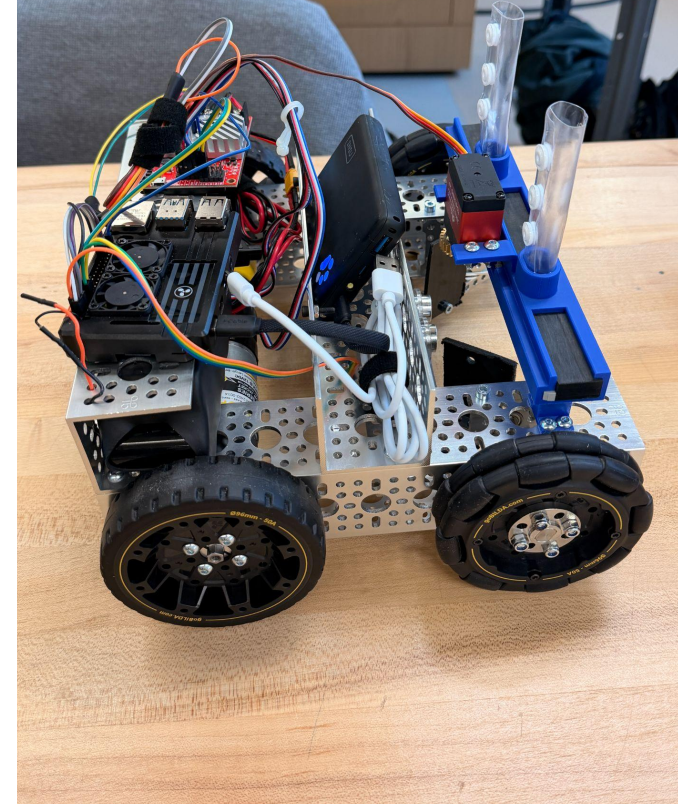
Has a rocker switch and a 20A fuse

Rechargeable 12V battery

MEDS Wiring Diagram



MEDS Final Circuit



Electronics Diagram

Approved by ISALL

2 Identical Wearable
Devices

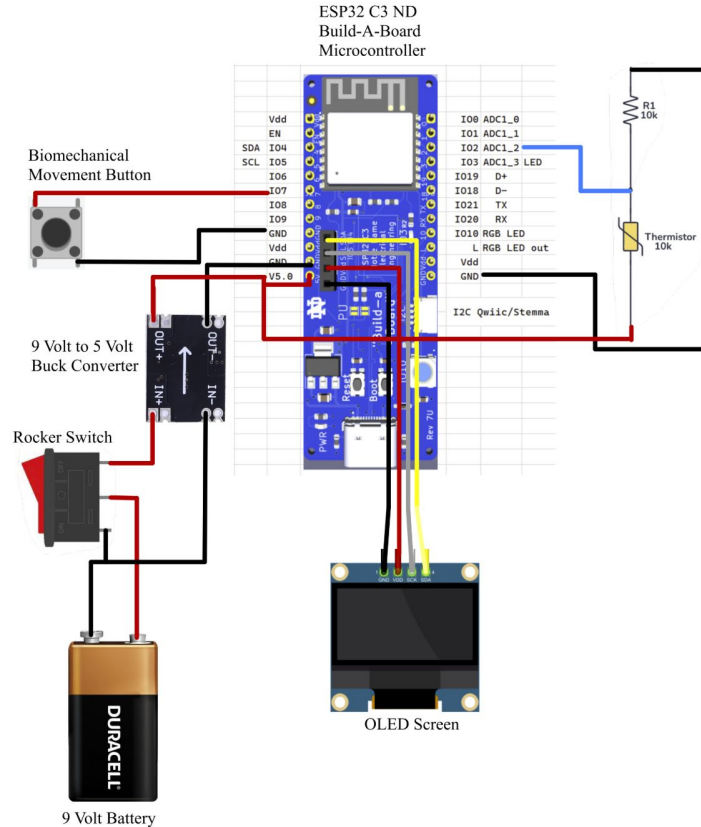
Microcontrollers:

- ESP32 C3 (x2)

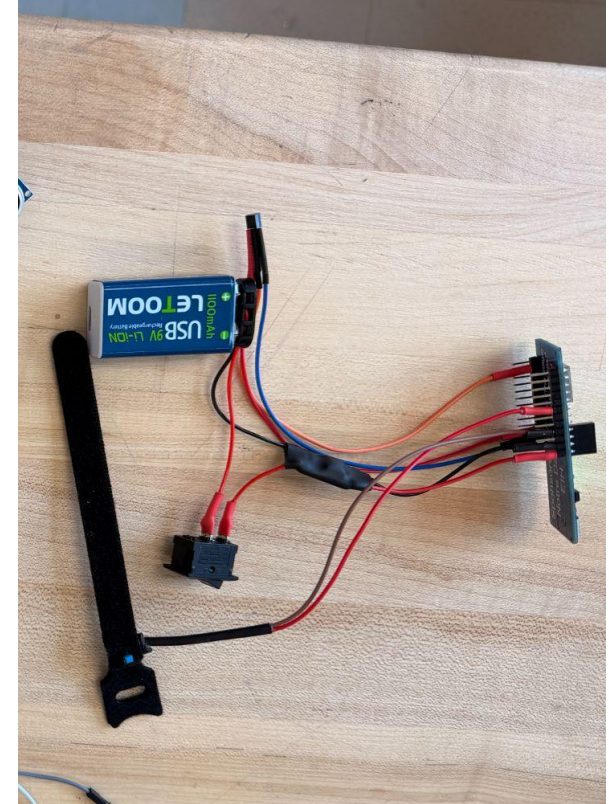
Both contain rocker
switches

Rechargeable 9V
battery stepped down to
5V

Wearable Device Wiring Diagram



Wearable Final Circuit



Changes Made to Electronics

- The servo is now powered using the 12v GoBuilda battery with a 12v to 5v buck converter
 - Changed to prevent Raspberry Pi brownouts due to large power draw
- The roboclaw motor controller is plugged into the portable battery bank to provide more stability to the 5v line
 - Done to mitigate loss of encoder signal errors due to insufficient power
- 1k and 2k resistors were added to the ultrasonic sensor wiring to ensure 3.3 volts were returned to the Raspberry Pi rather than 5v, preventing damage to the GPIO pins
- Adafruit BNO055 IMU sensor with 4.7k pull up resistor was added to improve 90 degree turn accuracy as compared with encoder values.
 - PI controller is used to control the motors

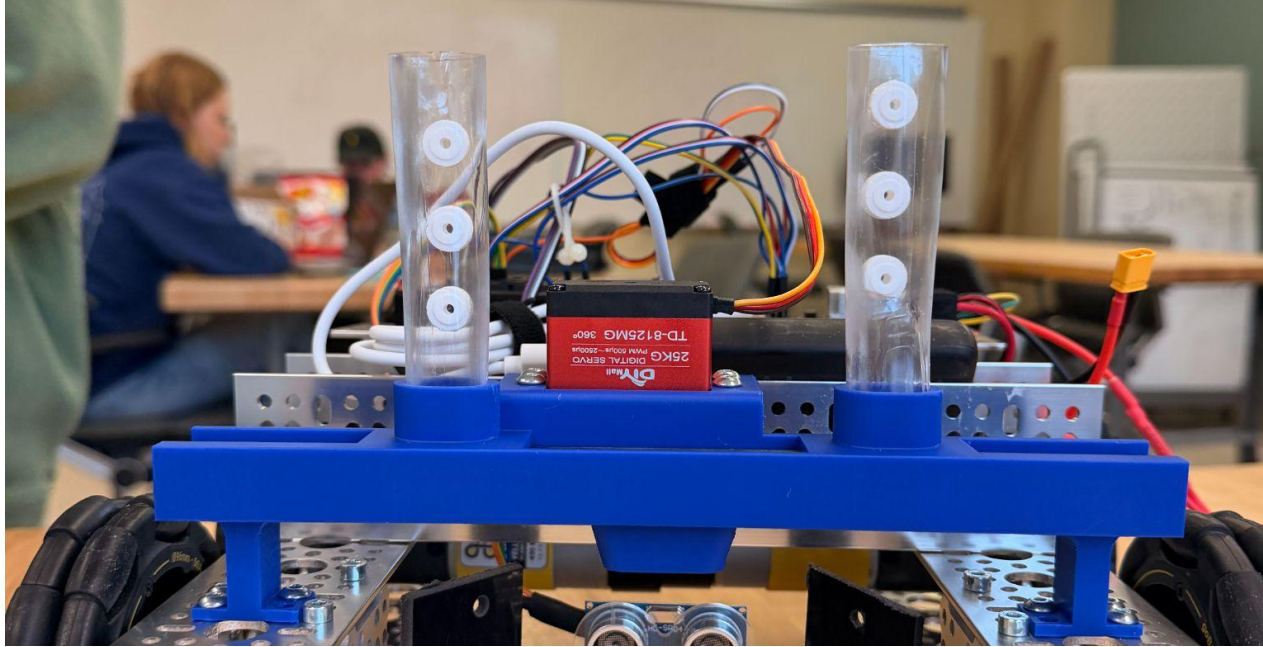
Programming / Code Highlights

- Drivetrain
 - Uses a PID control loop for both position and velocity via the roboclaw_3 python library
- The wearable device
 - Uses an 11-point median to determine an accurate and stable temperature reading
 - Connects to the “Group09Robot” wifi network with a static IP address and sends data via http
- The PiGPIO service is used to control the servo via hardware PWM
- Ultrasonic sensor is used for position validation and koozie searching via a swing turn
- IMU used to control robot orientation and turning accuracy via a PI controller
- OLED screen is used to display live patient status as well as robot odometry

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Pill Dispenser - Final Design

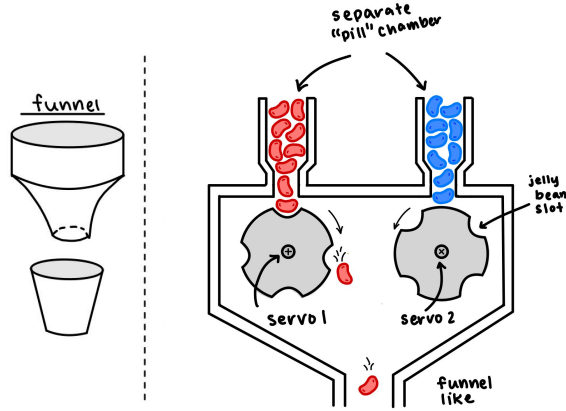


Pill Dispenser - Critical Design Decisions

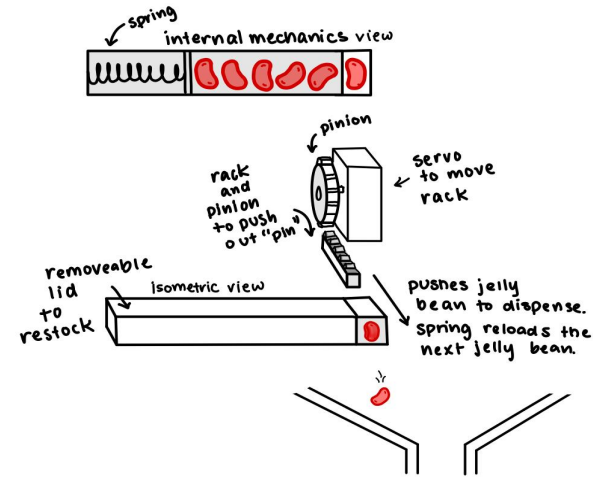
1. The subsystem should not require more than one servo
 - a. Reduces power consumption (6V)
2. Limited to one 3D print
3. Ensure only two jelly beans are dispensed (No Jamming)
 - a. 4.9677 in^3 volume for 20 jelly beans
4. Minimize cost

Pill Dispenser - Brainstorming

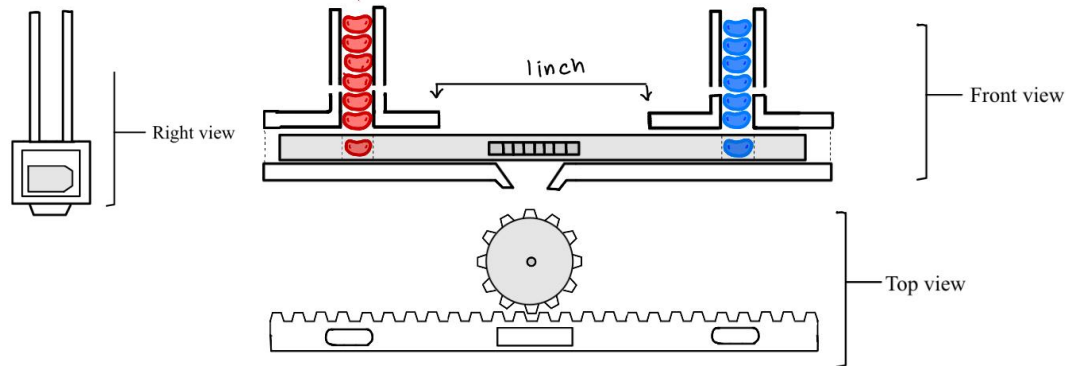
Concept 1



Concept 2



Final Concept



Pill Dispenser - Trade Study Results

- Design Variables:
 - Servo Motor
 - Pinion Gear
- State Variables:
 - Rack Force (45%)
 - Rack Velocity (35%)
 - Power Consumption (15%)
 - Cost (5%)
- Model
 - Matlab
 - Torque vs. Speed
 - Rack and pinion kinematics

DC linear approximation

$$T(\omega) = T_{stall} \left(1 - \frac{\omega}{\omega_{NL}} \right)$$

Angular Speed

$$\omega = \frac{2\pi}{T_{rev}} \quad (\text{rad/s})$$

$$\text{RPM} = \frac{60}{T_{rev}(s)}$$

Linear Rack Speed

$$v = \omega r = \frac{2\pi r}{T_{rev}} = \frac{C}{T_{rev}}$$

Power

$$P = IV = \tau\omega$$

Pill Dispenser - Trade Study Decision Matrix

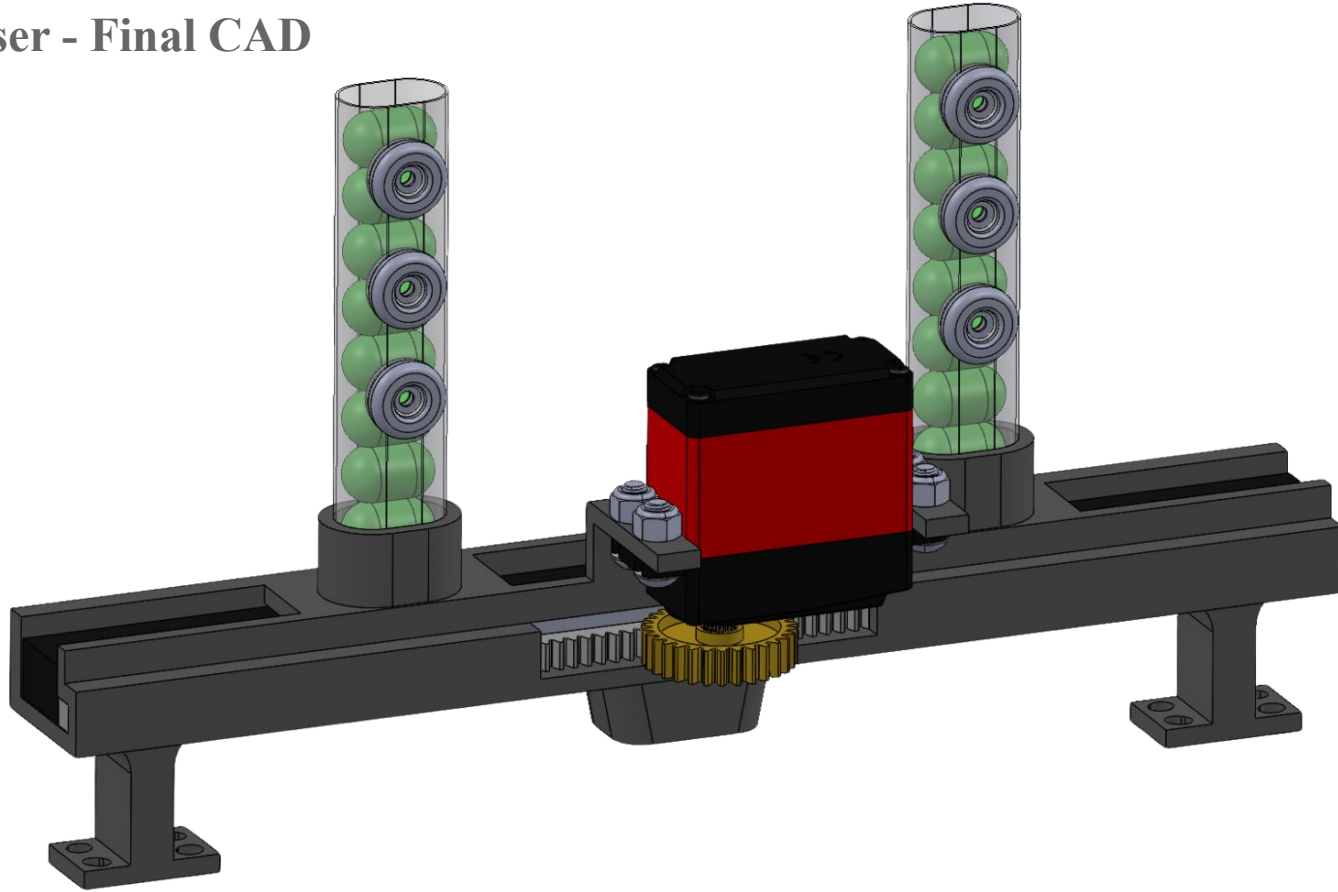
Find the best combination that satisfies requirements.	State Variable Results (1-10 Scale)								
	Rack Velocity		Applied Force from Rack		Power Consumption		Cost		Total
	0.45		0.35		0.15		0.05		1
Designs	Absolute	Weighted	Absolute	Weighted	Absolute	Weighted	Absolute	Weighted	Weighted
Gear 1 + Servo 1	0.00	0.00	10.00	3.50	10	1.50	10	0.50	5.50
Gear 1 + Servo 2	1.29	0.58	7.74	2.71	6.44	0.97	6.47	0.32	4.58
Gear 1 + Servo 3	3.40	1.53	3.09	1.08	1.42	0.21	1.18	0.06	2.88
Gear 1 + Servo 4	1.21	0.54	0.00	0.00	3.77	0.57	1.18	0.06	1.17
Gear 2 + Servo 1	0.67	0.30	8.73	3.06	8.86	1.33	9.41	0.47	5.16
Gear 2 + Servo 2	2.83	1.27	8.29	2.90	5.41	0.81	5.88	0.29	5.28
Gear 2 + Servo 3	6.34	2.85	5.44	1.90	0.47	0.07	0.59	0.03	4.86
Gear 2 + Servo 4	2.67	1.20	2.15	0.75	1.88	0.28	0.59	0.03	2.26
Gear 3 + Servo 1	1.49	0.67	8.76	3.07	8.31	1.25	8.82	0.44	5.42
Gear 3 + Servo 2	4.74	2.13	8.77	3.07	4.89	0.73	5.29	0.26	6.20
Gear 3 + Servo 3	10.00	4.50	6.85	2.40	0	0.00	0	0.00	6.90
Gear 3 + Servo 4	4.50	2.03	4.19	1.46	0.94	0.14	0	0.00	3.63

- Concept Recommendation
 - Gear 3, Motor 3: Continuous, bidirectional, controllable PWM (Highest RPM)
 - MOD 0.8, 30 Tooth Gear (Largest Gear)

Pill Dispenser - Design Changes

- Different servo
- Changes after Prototype Print 1:
 - Clearance between rack and 3D print changed to 0.005 inches.
- Changes after Prototype Print 2:
 - Servo mount on the 3D printed dimension changes and print orientation changed to increase strength.
 - Dispensing funnel increased by 1/10 inch.
- ABS-Bar jelly bean slot
 - Extra acetal plastic press-fit used to decrease jb slot size in ABS stock

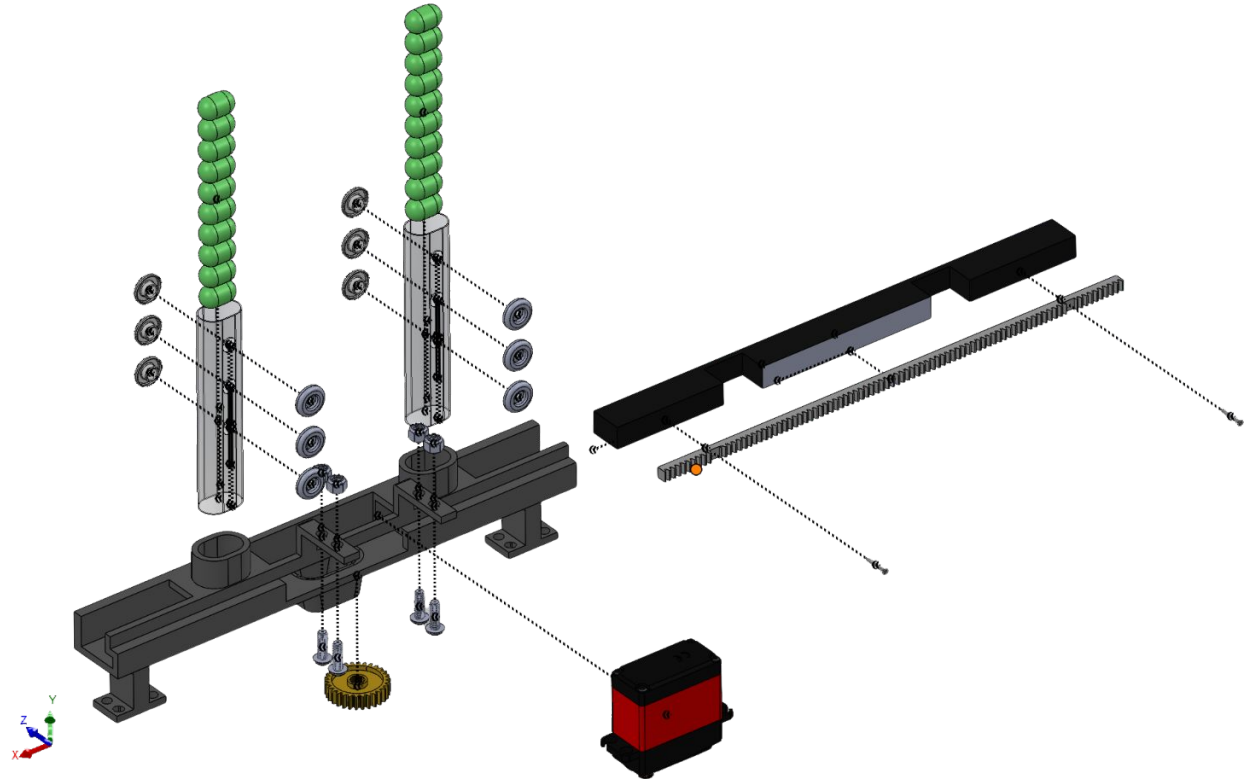
Pill Dispenser - Final CAD



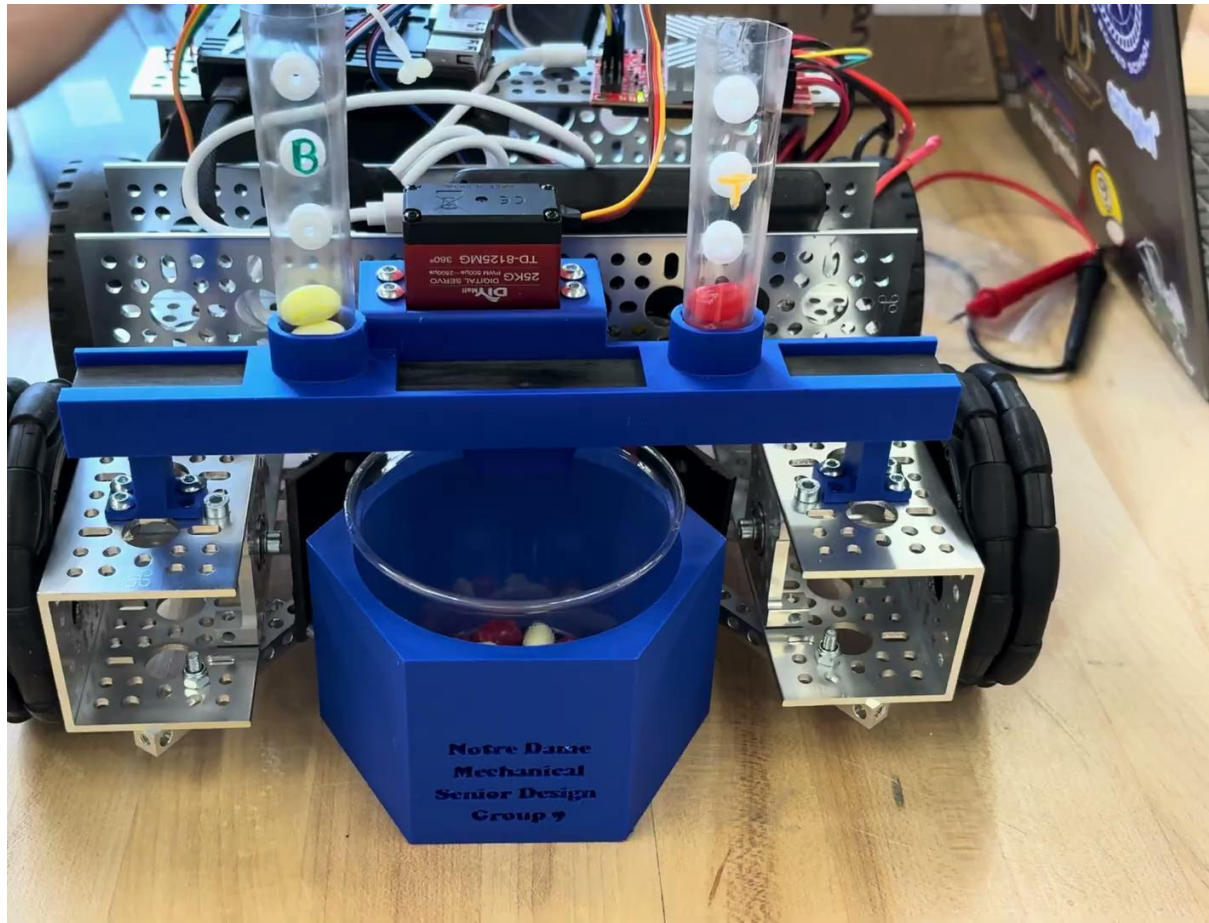
Pill Dispenser - Final Design

The pill dispenser had a total of 5 components that were subject to manufacturing.

- FDM printing (ABS-M30)
- Manual milling and drilling
- Water Jetting
- Dremel



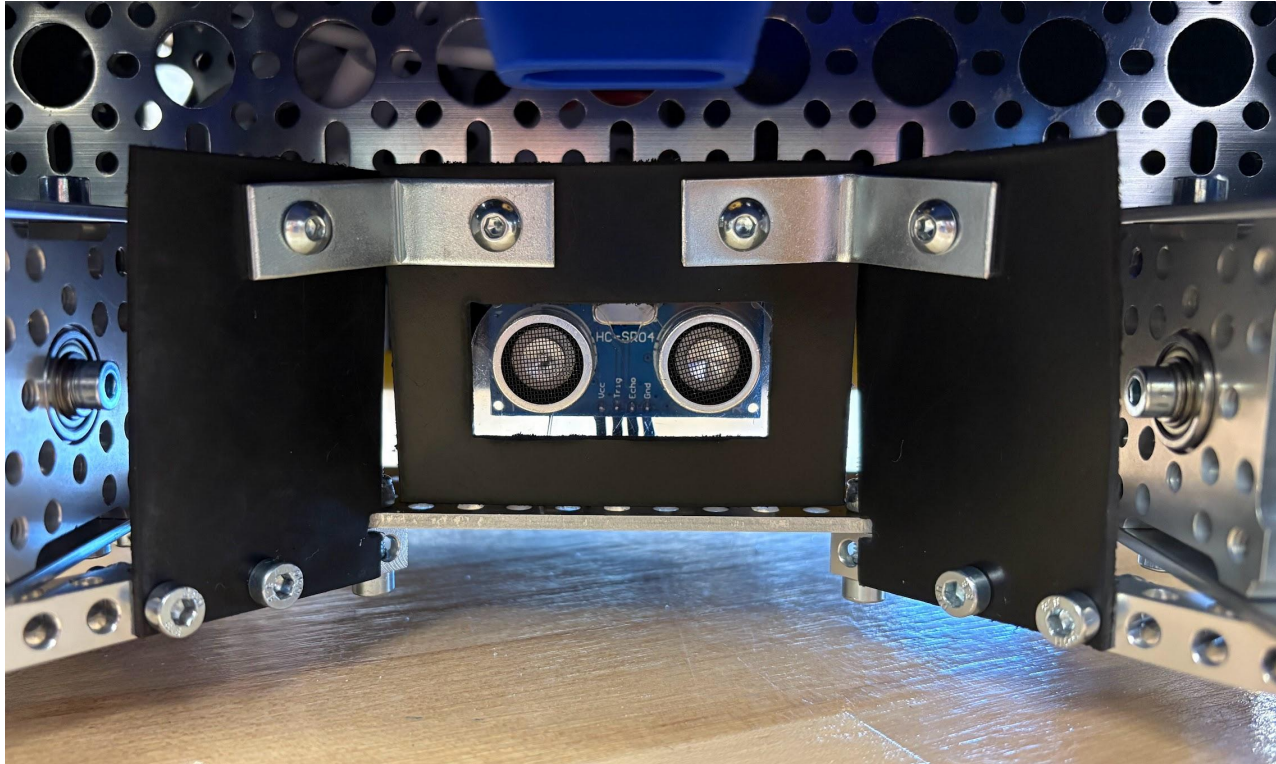
Pill Dispenser



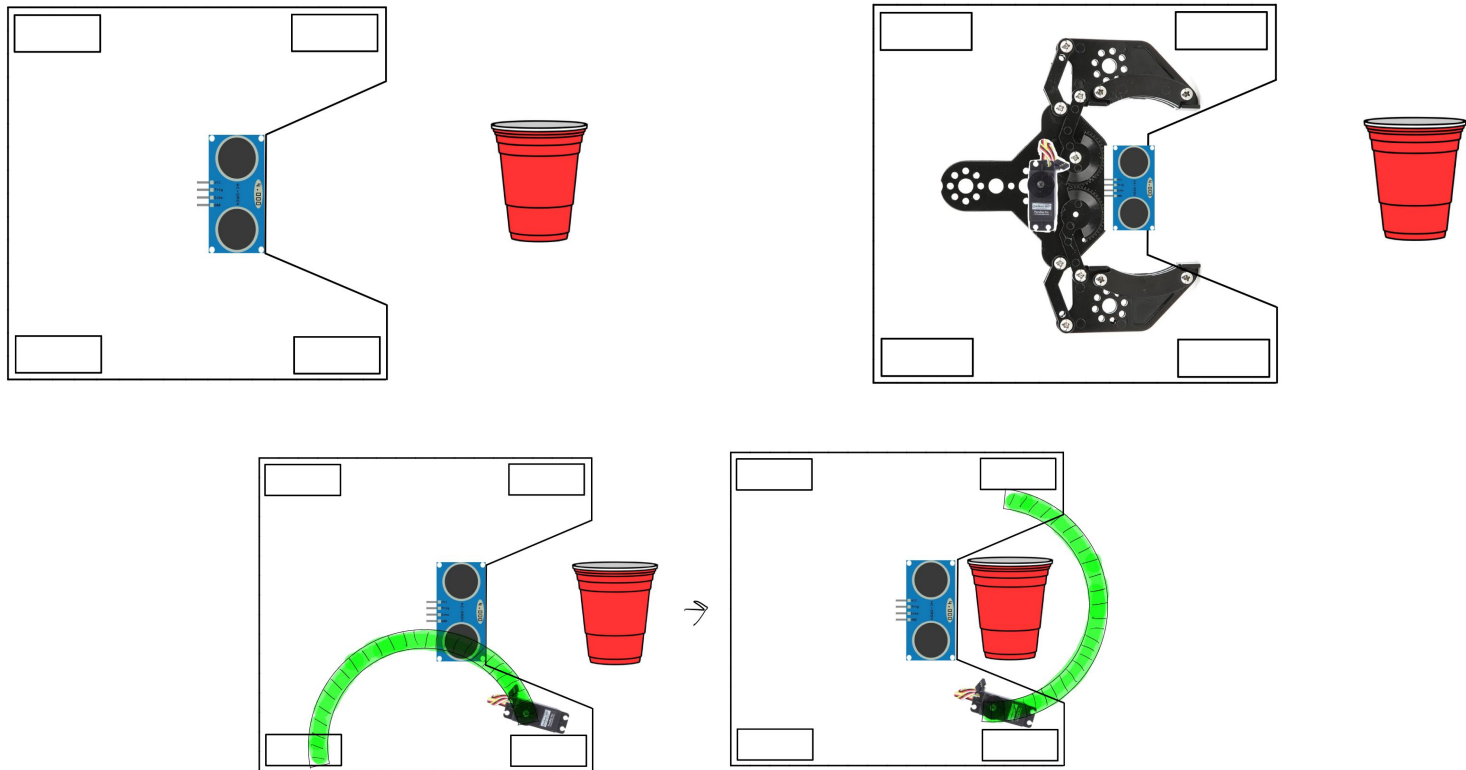
Agenda

- Project Overview
- Solution
- Build Process
- CAD
- Subsystem Breakdown
 - Chassis
 - Drivetrain
 - Electronics
 - Pill Dispenser
 - Cup Interface
 - Cover
 - Wearable
 - Koozie
- Budget
- Reflection

Cup Interface - Final Design



Cup Interface - Brainstorming



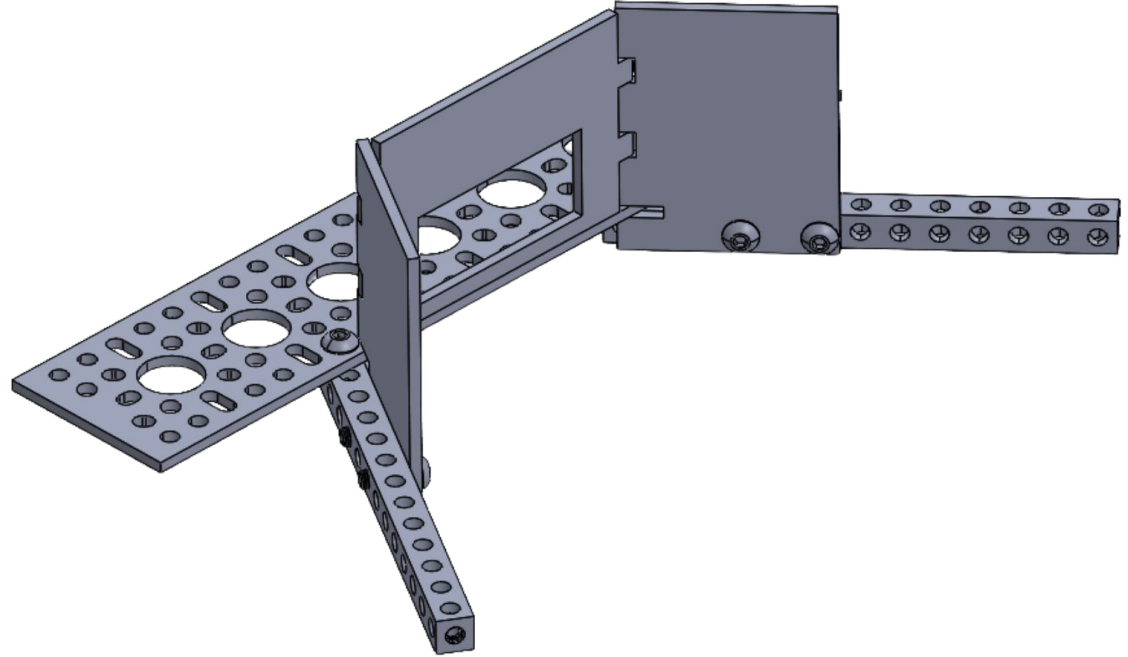
Cup Interface - Critical Design Decisions

Sensor

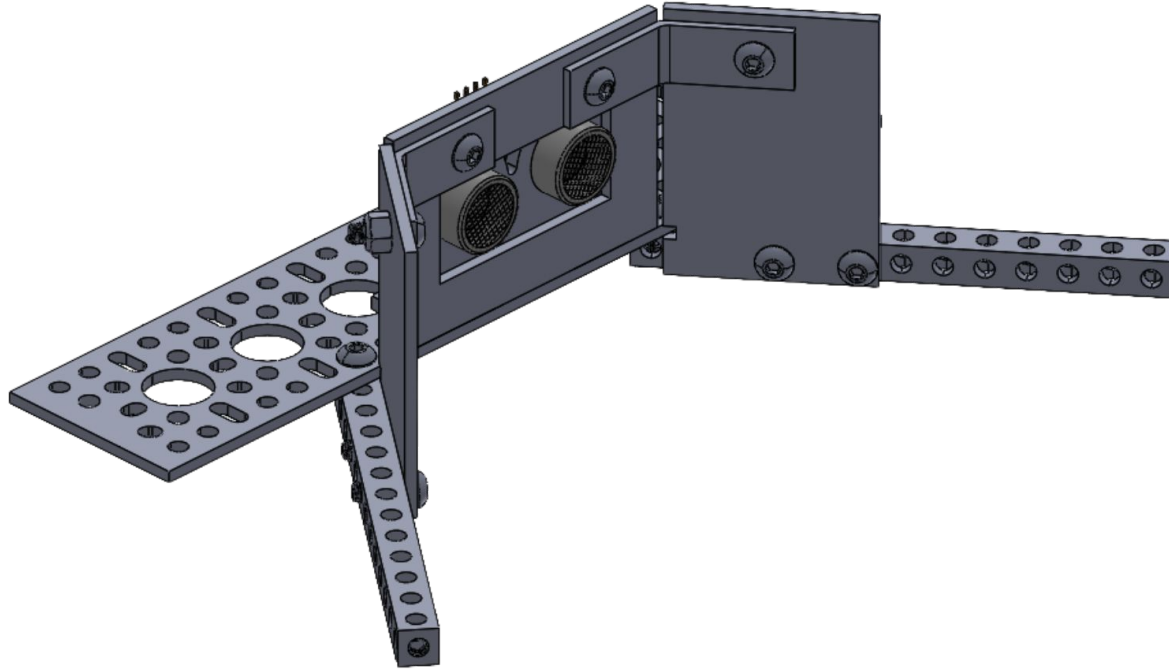
- Supplements autonomous driving by sensing the koozie
- Selected for ease of integration

Mesh with the koozie

- Angle must be 120°
- Must fit directly below pill chamber dispensing



Cup Interface - Final CAD



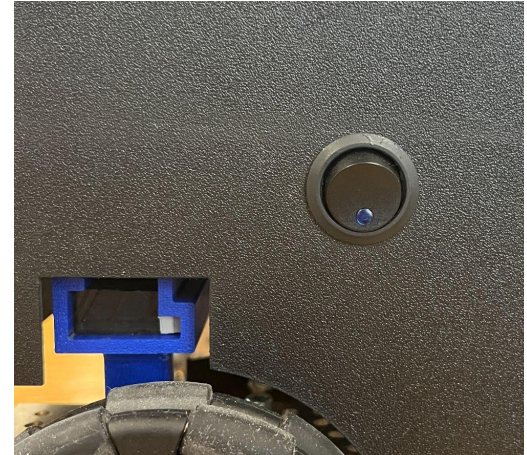
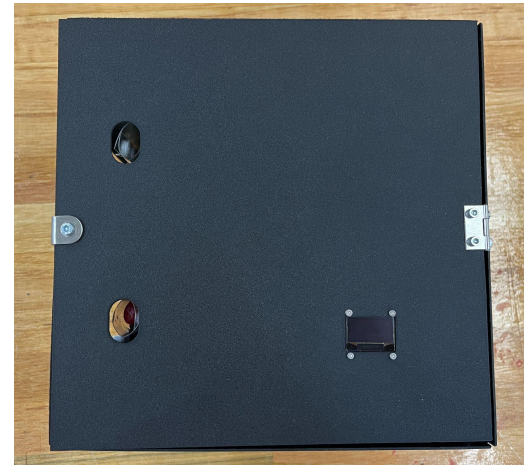
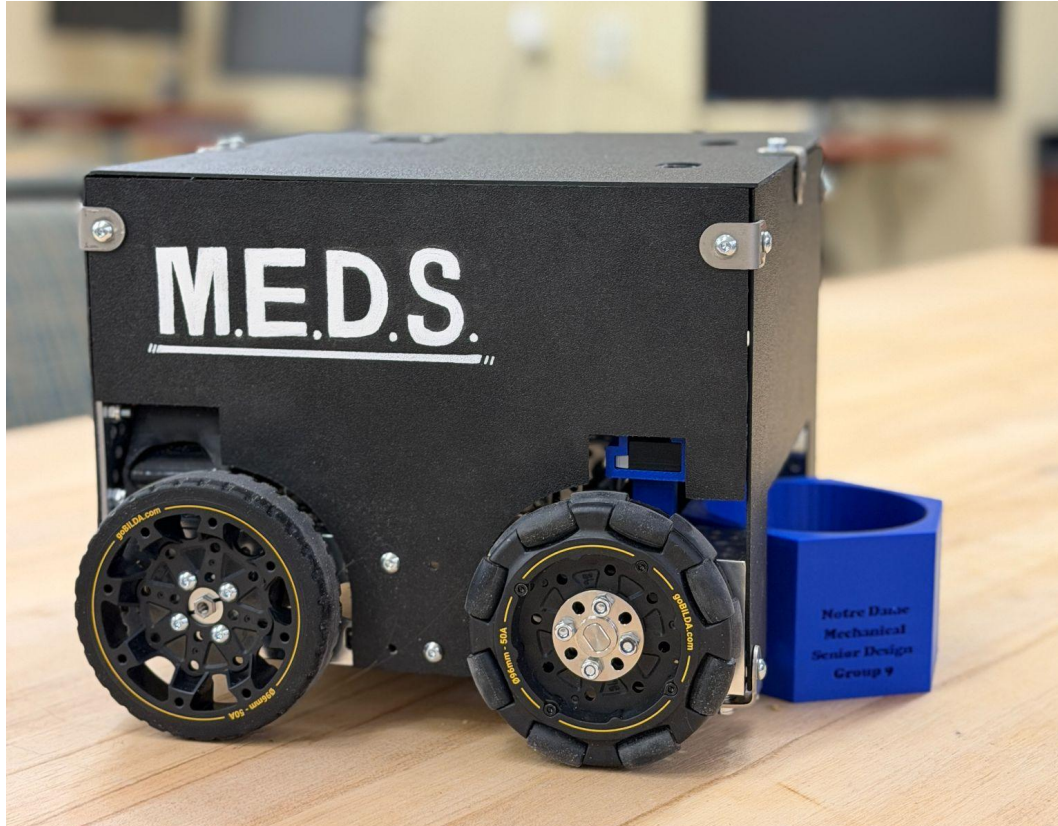
Design Changes

1. Swapped 1/8 inch HDPE with 1/16 inch ABS
 - a. Consistent with the covering of the robot
2. Bend 90° brackets to the desired angle rather than using a press fit

Agenda

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 - Koozie
- Budget
- Reflection

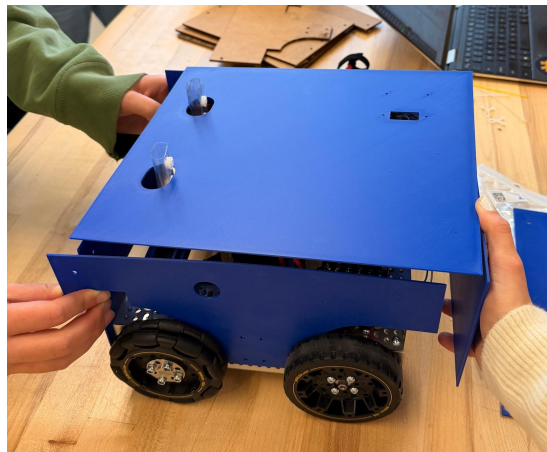
Covering- Final Design



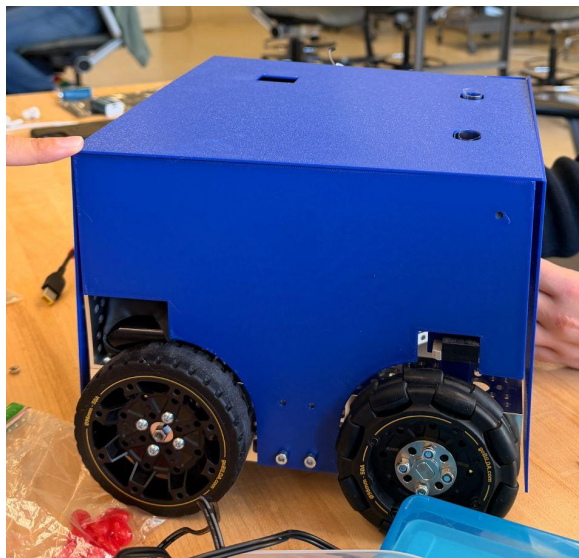
Cover- Design Changes

1 → 2: Updated length of U channels on the chassis

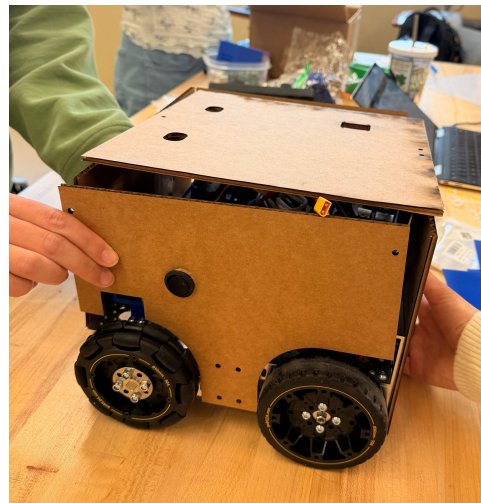
2 → 3: Updated location of fasteners



1

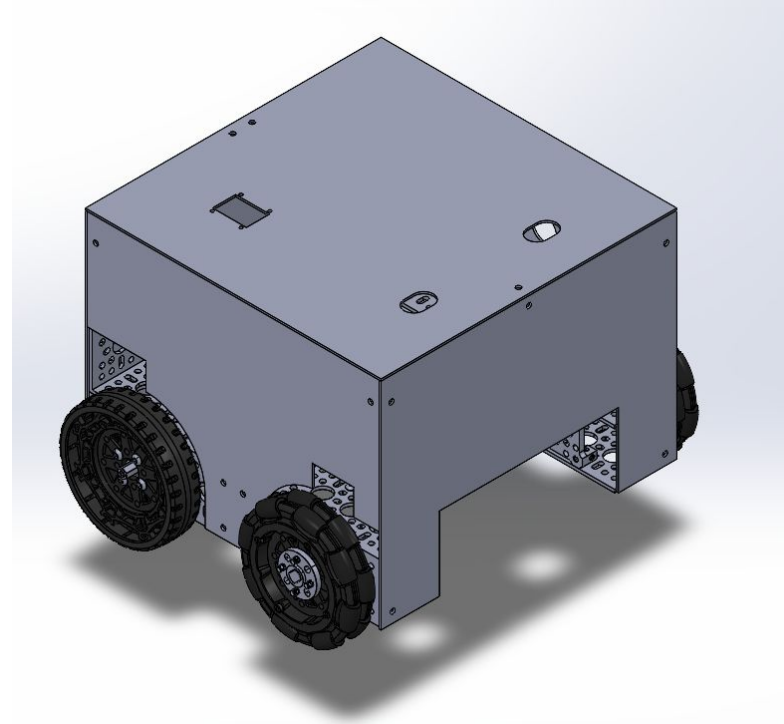
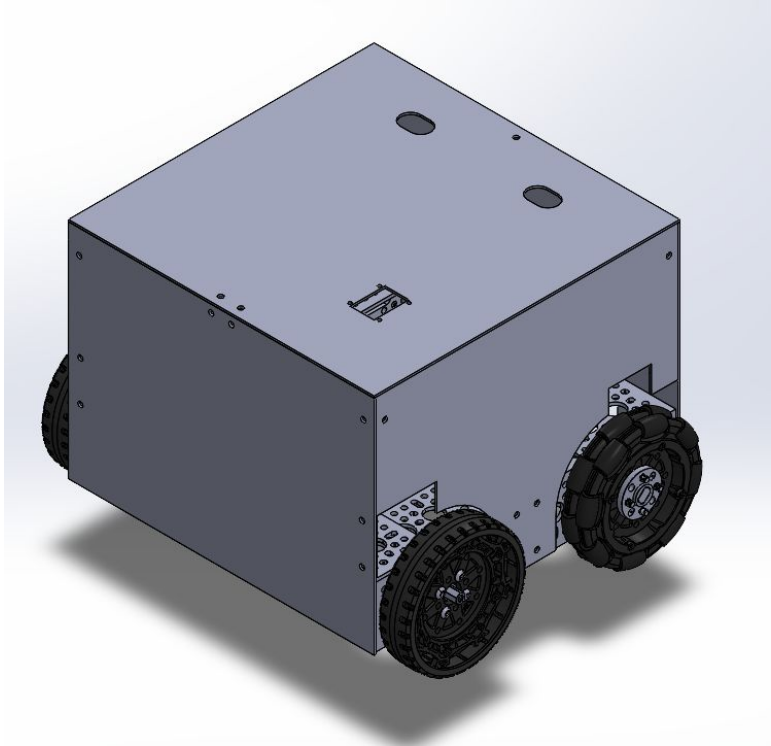


2



3

Cover- Final CAD

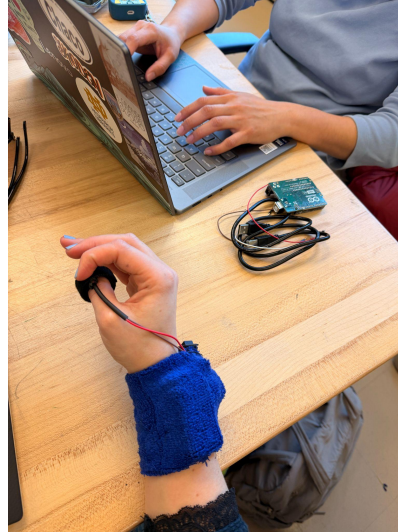


Agenda

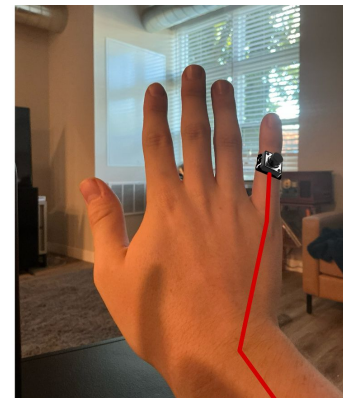
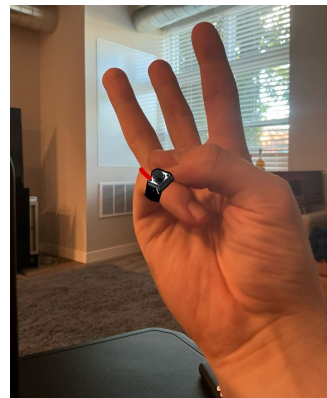
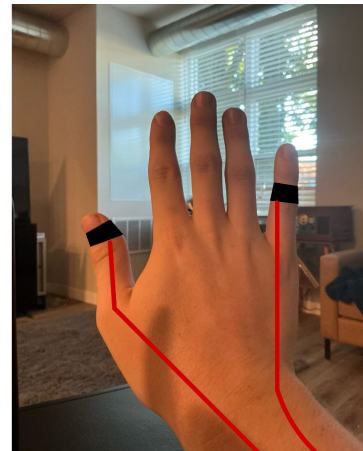
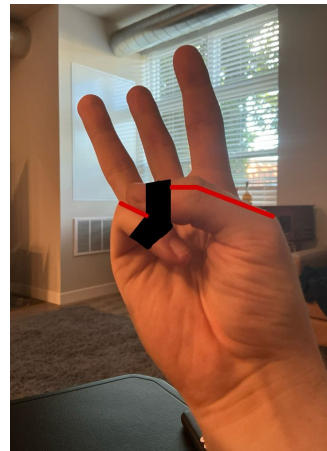
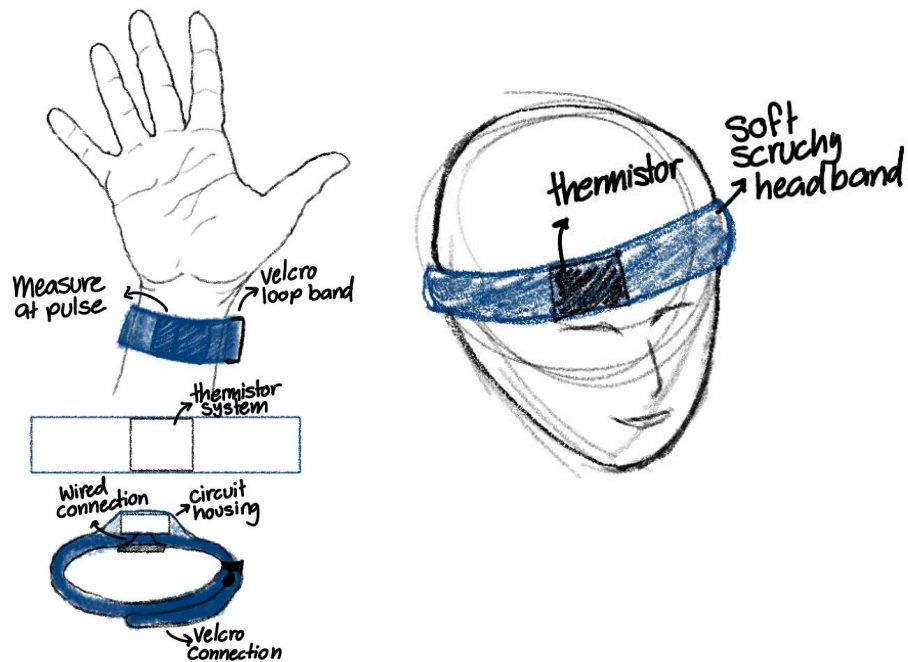
- Project Overview
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- Budget
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Wearable - Final Design

1. Was ranked an average of 0.93 on a Likert scale where 0 is extremely comfortable and 5 is extremely uncomfortable by three non-affiliated persons



Wearable - Brainstorming



Wearable - Critical Design Decisions

Thermal Sensor

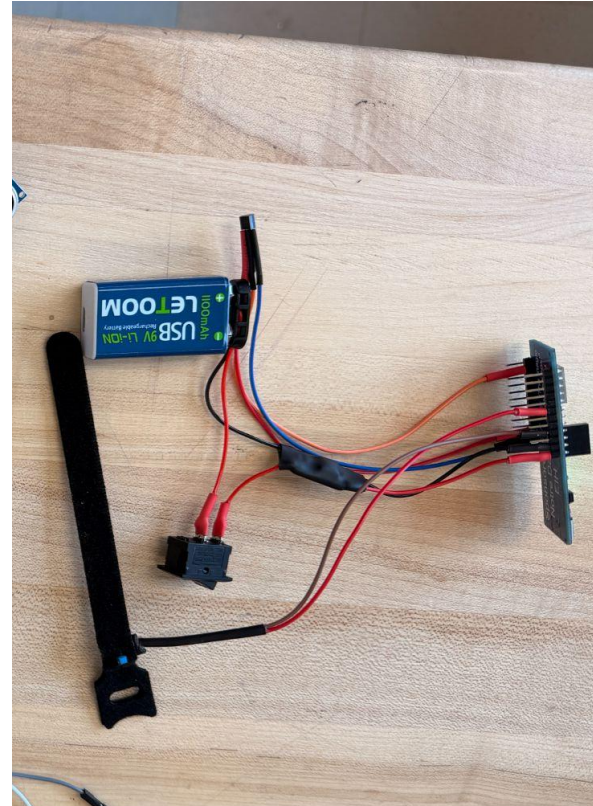
- Determined by trade study

Material

- Needed to be comfortable and able to support the electric circuit
- Originally chosen to be a nylon webbing strap

Biomechanical Movement Detection

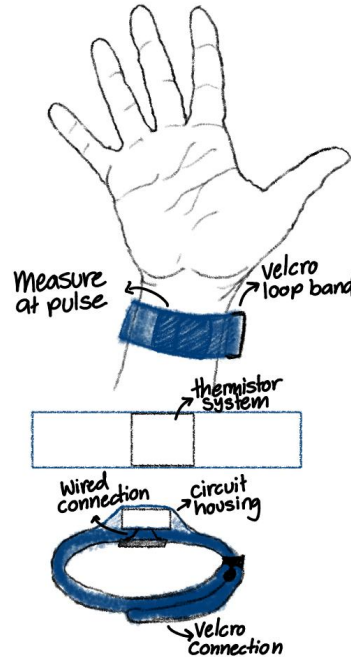
- Button was determined to be the ideal choice due to risks associated with an open circuit on the patient



Wearable - Trade Study Results

- Design Variables:
 - Contact Material
 - Contact Material Thickness
 - β Value of Thermistor
- State Variables:
 - Steady State Temperature (40%)
 - Temperature Tolerance of Thermistor (35%)
 - Time to 98% of SS Temperature (25%)
- Model
 - Matlab
 - Newton's Law of Cooling
 - Heat Diffusion equation
 - Steinhart-Hart equation

Concept Design



Critical Equations

Newton's Law of Cooling for
Convection & Fourier's Law

$$-k \frac{\partial T}{\partial x} \Big|_{x=L} = h[T(L, t) - T_{\infty}]$$

Steady State Temperature

Derived from above equation and boundary
conditions

$$T_{ss}(L) = T_{skin} - \frac{hL}{k + hL}(T_{skin} - T_{\infty})$$

Steinhart-Hart Equation

Simplified version

$$R(T) = R_0 * e^{\beta * (\frac{1}{T} - \frac{1}{T_0})}$$

Wearable - Trade Study Decision Matrix

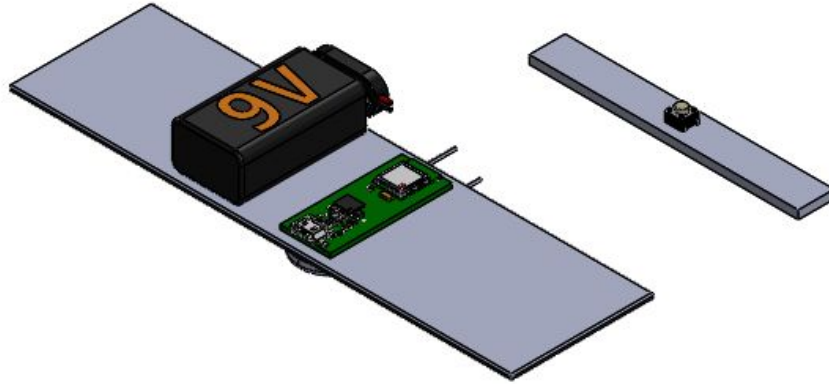
Determine the alternative that best meets the engineering specifications	State Variable Results (1-10 Scale)						
	Steady State Temperature		Temperature Tolerance		Time to 98% Accuracy		Total
	0.4		0.35		0.25		1
Designs	Absolute	Weighted	Absolute	Weighted	Absolute	Weighted	Weighted
Design 1	10.00	4.00	8.31	2.91	8.88	2.22	9.13
Design 2	10.00	4.00	1.81	0.63	8.88	2.22	6.85
Design 3	10.00	4.00	8.73	3.06	8.88	2.22	9.28
Design 4	10.00	4.00	1.00	0.35	8.88	2.22	6.57
Design 5	10.00	4.00	10.00	3.50	8.88	2.22	9.72
Design 6	2.87	1.15	8.31	2.91	10.00	2.50	6.56
Design 7	2.87	1.15	1.81	0.63	10.00	2.50	4.28
Design 8	2.87	1.15	8.73	3.06	10.00	2.50	6.70
Design 9	2.87	1.15	1.00	0.35	10.00	2.50	4.00
Design 10	2.87	1.15	10.00	3.50	10.00	2.50	7.15
Design 11	3.70	1.48	8.31	2.91	4.81	1.20	5.59
Design 12	3.70	1.48	1.81	0.63	4.81	1.20	3.32
Design 13	3.70	1.48	8.73	3.06	4.81	1.20	5.74
Design 14	3.70	1.48	1.00	0.35	4.81	1.20	3.03
Design 15	3.70	1.48	10.00	3.50	4.81	1.20	6.18
Design 16	1.00	0.40	8.31	2.91	1.00	0.25	3.56
Design 17	1.00	0.40	1.81	0.63	1.00	0.25	1.28
Design 18	1.00	0.40	8.73	3.06	1.00	0.25	3.71
Design 19	1.00	0.40	1.00	0.35	1.00	0.25	1.00
Design 20	1.00	0.40	10.00	3.50	1.00	0.25	4.15

- Results
 - Tradeoff between SS temperature and time of response for materials; the SS temperature was prioritized
 - Thermistors all perform relatively similarly to each other
- Concept Recommendation
 - EVA Foam
 - Thickness: 0.079 inches
 - β Value of Thermistor: 4110 K

Wearable - Design Changes

1. Originally the electronics were going to be secured to a wristband made of nylon webbing. However when the board was selected for the wearable, it was too long for standard webbing widths. The team pivoted to a stretchy sweat band which was:
 - a. Wider: allowing for all of the electronics to be accommodated
 - b. Stretchy: allowing it to fit many different patients
 - c. Hollow: allowing the electronics to be secured on the inside and covered
2. Which finger should the button be placed on for the biomechanical movement
 - a. Thumb was chosen for ease of use by patient
3. How to ensure that the button is accidentally triggered
 - a. Button must be pressed 3 times

Wearable - Final CAD



Design Changes

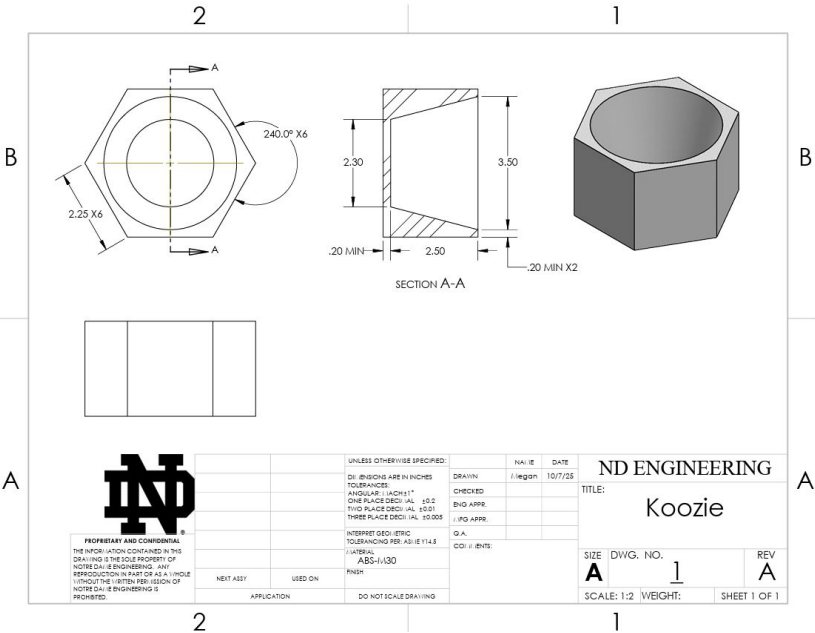
1. Place electronics inside the wristband rather than on top

* Difficult to CAD the flexible nature and circular shape of the wearable, modeled as a flat strip

Agenda

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 - Koozie
- Budget
- Reflection

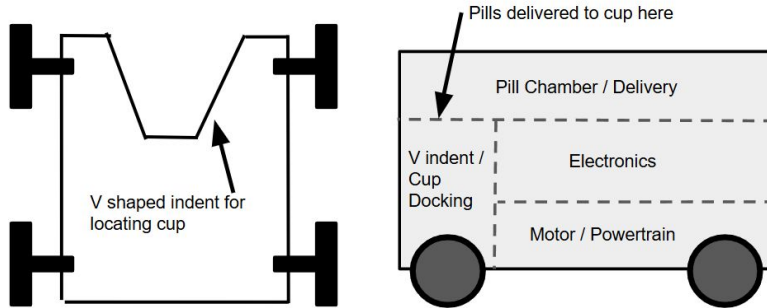
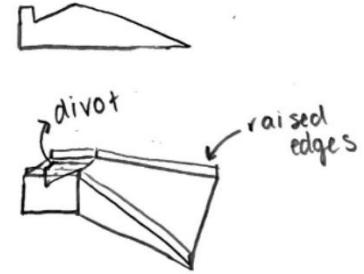
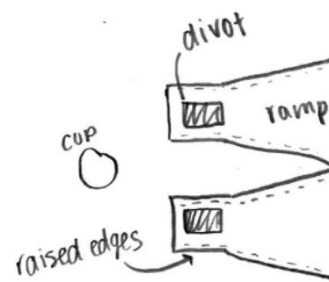
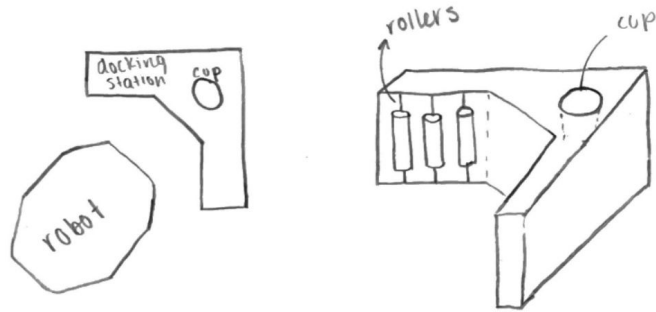
Koozie - Final Design



- Tapered Design
- Increases Cup security
- Lowers COM which allows for increased height without increased tipping probability

Koozie - Brainstorming

1. Delivery Device



- The final Koozie design was selected to mesh with this chassis design, leading to the “koozie” shape and hexagonal base

Koozie - Trade Study Results

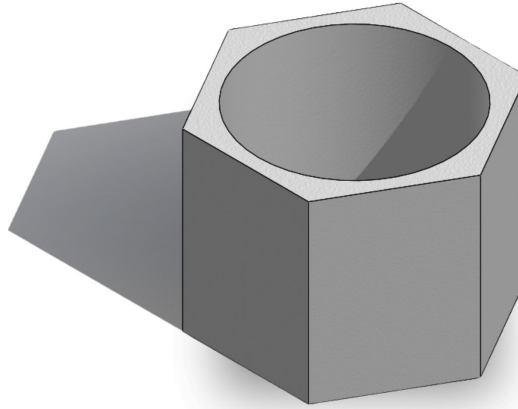
- Design Variables:
 - Material
 - Speed of MEDs at contact
 - Height of the Koozie
- State Variables:
 - Tipping Ratio (40%)
 - Sliding Distance (35%)
 - Time of interaction (20%)
 - Cost (5%)
- Model
 - Matlab code
 - Conservation of momentum
 - Work equals change in kinetic energy
 - Kinematics

$$d = \frac{v_i^2}{2\mu g}$$

$$R_{tip} = \frac{F_M(h - 1.5)(1.5 + \frac{h-1.5}{2})}{3.897F_g}$$

$$t_{interaction} = \frac{12}{v_o} + \frac{1}{v_2} + \frac{v_f - v_2}{\mu g}$$

Concept Design



Koozie - Trade Study Results Continued

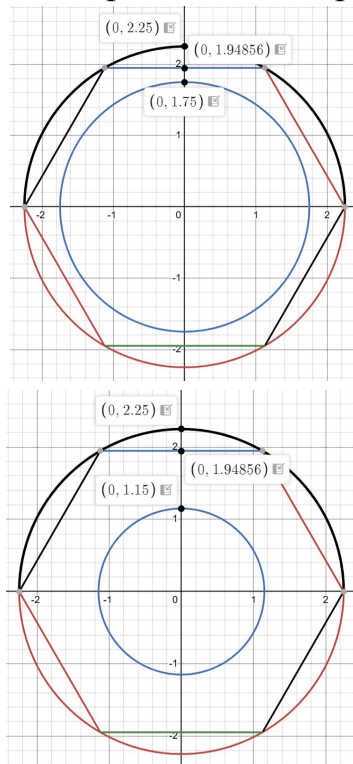
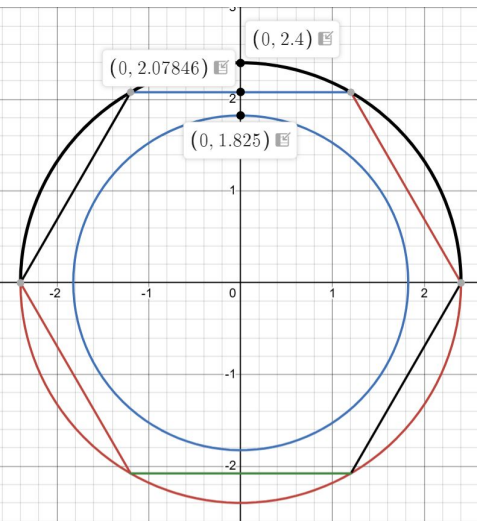
Determine the design of the Koozie that best meets the engineering specifications			State Variable Results (1-10 Scale)									
			Tipping Ratio		Disance Traveled after Contact		Time of Interaction		Cost		Total	
			0.4		0.35		0.2		0.05		1	
Designs												
Material	Height	Initial Velocity	Absolute	Weighted	Absolute	Weighted	Absolute	Weighted	Absolute	Weighted	Absolute	Weighted
HDPE	1.75	0.5	9.91	3.96	7.45	2.61	1.00	0.20	4.46	0.22	6.99	6.99
HDPE	1.75	0.75	9.79	3.92	4.27	1.49	6.60	1.32	4.46	0.22	6.95	6.95
HDPE	2	0.5	9.80	3.92	7.45	2.61	1.00	0.20	3.77	0.19	6.92	6.92
HDPE	2	0.75	9.55	3.82	4.27	1.50	6.60	1.32	3.77	0.19	6.82	6.82
HDPE	2.25	0.5	9.68	3.87	7.46	2.61	1.00	0.20	3.08	0.15	6.84	6.84
HDPE	2.25	0.75	9.28	3.71	4.28	1.50	6.60	1.32	3.08	0.15	6.68	6.68
HDPE	2.5	0.5	9.54	3.82	7.46	2.61	1.00	0.20	2.38	0.12	6.75	6.75
HDPE	2.5	0.75	8.97	3.59	4.28	1.50	6.60	1.32	2.38	0.12	6.53	6.53
HDPE	2.75	0.5	9.39	3.76	7.46	2.61	1.00	0.20	1.69	0.08	6.66	6.66
HDPE	2.75	0.75	8.64	3.46	4.29	1.50	6.61	1.32	1.69	0.08	6.36	6.36
HDPE	3	0.5	9.23	3.69	7.47	2.61	1.00	0.20	1.00	0.05	6.56	6.56
HDPE	3	0.75	8.27	3.31	4.30	1.50	6.61	1.32	1.00	0.05	6.18	6.18
ABS-30	1.75	0.5	9.91	3.96	8.28	2.90	1.48	0.30	7.42	0.37	7.53	7.53
ABS-30	1.75	0.75	9.79	3.92	6.14	2.15	7.32	1.46	7.42	0.37	7.90	7.90
ABS-30	1.75	1	9.62	3.85	3.14	1.10	10.00	2.00	7.42	0.37	7.32	7.32
ABS-30	2	0.5	9.80	3.92	8.28	2.90	1.48	0.30	7.17	0.36	7.47	7.47
ABS-30	2	0.75	9.54	3.82	6.14	2.15	7.32	1.46	7.17	0.36	7.79	7.79
ABS-30	2	1	9.19	3.68	3.14	1.10	10.00	2.00	7.17	0.36	7.13	7.13
ABS-30	2.25	0.5	9.68	3.87	8.28	2.90	1.48	0.30	6.91	0.35	7.41	7.41
ABS-30	2.25	0.75	9.27	3.71	6.14	2.15	7.32	1.46	6.91	0.35	7.67	7.67
ABS-30	2.25	1	8.70	3.48	3.14	1.10	10.00	2.00	6.91	0.35	6.92	6.92
ABS-30	2.5	0.5	9.54	3.82	8.29	2.90	1.48	0.30	6.66	0.33	7.34	7.34
ABS-30	2.5	0.75	8.96	3.58	6.14	2.15	7.32	1.46	6.66	0.33	7.53	7.53
ABS-30	2.5	1	8.15	3.26	3.14	1.10	10.00	2.00	6.66	0.33	6.69	6.69
ABS-30	2.75	0.5	9.39	3.75	8.29	2.90	1.48	0.30	6.41	0.32	7.27	7.27
ABS-30	2.75	0.75	8.62	3.45	6.14	2.15	7.32	1.46	6.41	0.32	7.38	7.38
ABS-30	2.75	1	7.55	3.02	3.14	1.10	10.00	2.00	6.41	0.32	6.44	6.44
ABS-30	3	0.5	9.22	3.69	8.29	2.90	1.48	0.30	6.15	0.31	7.19	7.19
ABS-30	3	0.75	8.25	3.30	6.14	2.15	7.32	1.46	6.15	0.31	7.22	7.22
ABS-30	3	1	6.88	2.75	3.14	1.10	10.00	2.00	6.15	0.31	6.16	6.16

- Results
 - Conclusive results for the material to be ABS-M30 and the speed of the MEDS to be 0.75 mph. With these variables set, the height can range from 1.75 in to 2.25 in with minimal impact
- Concept Recommendation
 - ABS-M30 3D printed
 - MEDS velocity of 0.75 mph
 - Height of 1.75
 - Can be increased depending on the final height of the chassis.

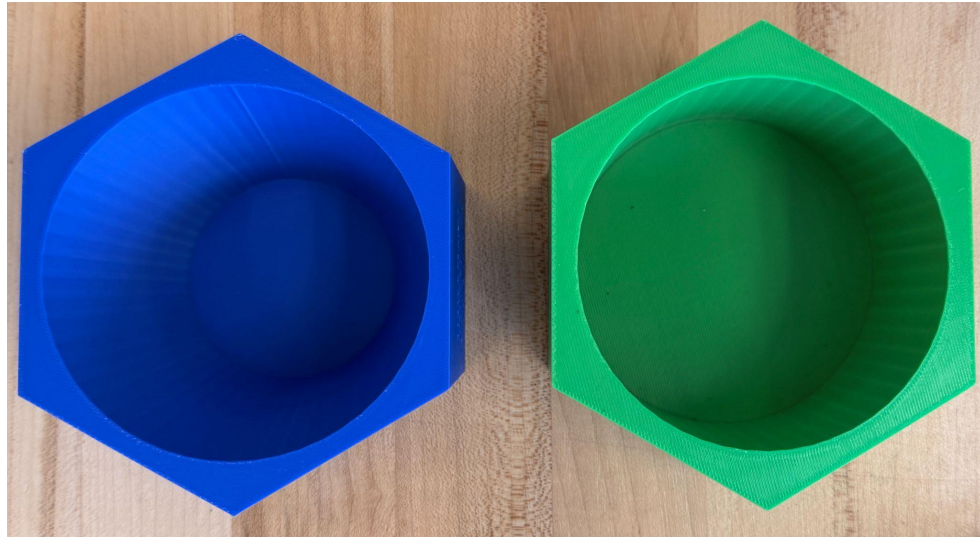
Koozie - Design Changes

Final Top and Base Shape

Original Base Shape



Design Comparison



Tapered Design

Straight Design

Agenda

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 - Koozie
- Budget
- Reflection

Budget Overview

Starting Amount		\$1,000.00
Amount Remaining on Card		\$307.38
Amount Remaining in Budget		\$163.68
Subsystem Breakdown	Drivetrain	\$534.96
	Chassis	\$101.74
	Pill Dispenser	\$80.62
	Koozie	\$11.64
	Wearable Devices	\$88.99
	Shipping	\$18.19
Total Spent		\$836.14

Agenda

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Reflections

- If we had two more weeks we would...
 - Add more sensors, such as having two ultrasonic sensors for finding the koozie
 - Create a better method for setting up the robot and ensuring everything was placed correctly
 - 3D print the chambers as part of the pill dispenser
 - Create a more complex cover design, possibly star wars inspired
- What would we do differently?
 - Higher quality thermistors
 - Design the Koozie so that it doesn't have "stealth mode" due to the angles
 - Prioritize bolt access when designing cover for ease of manufacturing
- Success
 - The robot was more successful than the team originally thought it would be
 - Aesthetics were not considered a priority at the beginning of the semester but we ended up having time to iterate through the cover design to improve the fit
 - We were able to devote time and resources to perfecting the course path and increasing the speed of the delivery robot

Thank You!

Questions?

Budget - Part 1

Account and Discount Information

Vendor Name	Username	Password					
GoBuilda	dshunk@nd.edu	Group09Senior Design\$				Starting Amount	\$1,000.00
Adafruit	Group09 or dshunk@nd.edu	Group09Senior Design\$				Amount Remaining on Card	\$307.38
DigiKey	dshunk@nd.edu	Group09Senior Design\$				Amount Remaining in Budget	\$163.86

Preliminary BOM

Subassembly	Name of part	Amount Required	Total Cost	Cost If Card Not Used (Components from ElH, ISALL, etc.)	Date Ordered	Receipt Link	Delivered/Obtained?
Drivetrain	435 RPM GoBuilda Yellow Jacket Motors	2	\$93.48	\$0.00	9/30	https://drive.google.com/drive/f	Yes
	1309 Series Sonic Hub (8mm REX™ Bore)	2	\$13.58	\$0.00	9/30	https://drive.google.com/drive/f	Yes
	Hogback Traction Wheels (Back Wheels)	2	\$16.98	\$0.00	9/30	https://drive.google.com/drive/f	Yes
	RoboClaw Motor Controller	1	\$84.99	\$0.00	9/30	https://drive.google.com/drive/f	Yes
	2802 Series Zinc-Plated Steel Button Head Screw (M4 x 0.7mm, 12mm Length) - 25 Pack	2	\$6.10	\$0.00	9/30	https://drive.google.com/drive/f	Yes
	96mm Omni Wheel (14mm Bore, 50A Durometer)	2	\$0.00	\$37.38	9/30	ElH	Yes
	Raspberry Pi 4B	1	\$59.99	\$0.00	10/2	https://drive.google.com/file/d/1	Yes
	12V NiMH Nested Battery (3000mAh, MH-FC XT30 Connector)	1	\$55.24	\$0.00	10/2	https://drive.google.com/file/d/1	No
	SanDisk 64GB Ultra microSDXC UHS-I Memory Card with Adapter - Up to 140MB/s, C10, U1, Full HD, A1, MicroSD Card - SDSQUAB-064G-GN6MA [New Version]	1	\$9.82	\$0.00	10/2	https://drive.google.com/file/d/1	Yes

Budget - Part 2

	Encoder Breakout Cable (4-Pos JST XH [MH-FC] to 4 x 1-Pos TJC8 [MH-FC], 300mm Length)	2	\$6.78	\$0.00	10/2	https://drive.google.com/file/d/1...	Yes
	6mm 1611 Series Flanged Ball Bearing	1	\$3.39	\$0.00	9/30	https://drive.google.com/drive/f...	Yes
	8mm 1611 Series Flanged Ball Bearing	1	\$3.39	\$0.00	10/2	https://drive.google.com/file/d/1...	Yes
	1201 Series Quad Block Pattern Mount (43-2)	2	\$11.88	\$0.00	9/30	https://drive.google.com/drive/f...	Yes
	Shouldered 8mm REX™ Hub-Shaft (20mm Hub Offset)	2	\$22.08	\$0.00	9/30	https://drive.google.com/drive/f...	Yes
	Sheet Material for covering	1	\$17.99	\$0.00	10/28	https://www.amazon.com/gp/pr...	Yes
	5V battery pack	1	\$18.69	\$0.00	10/9	https://drive.google.com/file/d/1...	Yes
	USB - Micro USB	1	\$4.99	\$0.00	11/4	https://www.amazon.com/gp/pr...	Yes
	Laser Pointer	1	\$9.49	\$0.00	11/4	https://www.amazon.com/gp/pr...	Yes
	BNO055 IMU Sensor	1	\$0.00	\$34.95	11/24	ISAL	Yes
	White paints pens	1	\$7.99	\$0.00	11/25	amazon.com/gp/product/B0D2...	Yes
	Acrylic Sheet	1	\$15.78	\$0.00	11/25	https://www.amazon.com/gp/pr...	Yes
Chassis	1120 Series U-Channel (9 Hole, 240mm Length)	4	\$44.16	\$0.00	9/30	https://drive.google.com/drive/f...	Yes
	M4 x 0.7mm Zinc-Plated Socket Head Screw (8mm Length) (25 Pack)	2	\$5.42	\$0.00	9/30	https://drive.google.com/drive/f...	Yes
	M4 x 0.7mm Nylock Nut (25 Pack)	2	\$5.08	\$0.00	9/30	https://drive.google.com/drive/f...	Yes
	1123 Series Pattern Plate (1 x 7 Hole, 43 x 192)	1	\$5.09	\$0.00	9/30	https://drive.google.com/drive/f...	Yes
	1106 Series Square Beam (13 Hole, 104mm L)	2	\$8.14	\$0.00	9/30	https://drive.google.com/drive/f...	Yes
	5 Pcs HC-SR04 Ultrasonic Sensor	1	\$8.99	\$0.00	10/2	https://drive.google.com/file/d/1...	Yes
	Ultrasonic Sensor brackets	1	\$6.99	\$0.00	10/9	https://drive.google.com/file/d/1...	Yes
	1/8 Inch Thick, 6" x 12" High Density Polyethyl	1	\$12.99	\$0.00	10/2	https://drive.google.com/file/d/1...	Yes
	Hinge	1	\$0.00	\$0.85	11/6	EIH	Yes
	Copper Brass 25mm x 25mm 60 degree	7	\$0.00	\$0.45	11/6	EIH	Yes

Budget - Part 3

	Corner Braces 25mm x 25mm 90 degree	7	\$0.00	\$3.15	11/6	ElH	Yes
	1 in. Zinc-Plated Corner Brace	2	\$0.00	\$0.88	10/9	ElH	Yes
Pill Chamber	TD-8125MG 25KG Digital Servo 360 Degree	1	\$22.88	\$0.00	10/2	https://drive.google.com/file/d/1	Yes
	2305 Series Brass, MOD 0.8, Servo Gear (25 Tooth Spline, 30 Tooth)	1	\$9.34	\$0.00	9/30	https://drive.google.com/drive/f	Yes
	20 Degree Pressure Angle Gear Rack, 32 Pitch (1 ft)	1	\$3.87	\$3.87	10/2	https://drive.google.com/file/d/1	Yes
	3D Printing of Housing with ABS-M30	1	\$0.00	\$18.34			Yes
	ABS Bar Stock (1 ft Rectangular Stock)	1	\$4.80	\$4.80	10/2	https://drive.google.com/file/d/1	Yes
	#0-80 x 1/2" Phillips Drive Stainless Steel Flat Head Machine Screw	2	\$0.00	\$1.18	Pre owned	N/A	Yes
	Plastic Snap Fasteners	6	\$0.00	\$3.00	Pre owned	N/A	Yes
	Jelly bean chambers	1	\$8.54	\$0.00	10/2	https://drive.google.com/file/d/1	Yes
Koozie	3D printing with ABS-M30	2	\$0.00	\$11.64			Yes
Wearable Device	Thermistors	1	\$13.99	\$0.00	11/4	https://www.amazon.com/gp/pr	Yes
	Button	2	\$0.00	\$0.20	10/9	ISALL	Yes
	Wiring	1	\$7.99	\$3.28	10/9	ISALL and https://drive.google	Yes
	Velcro Straps	1	\$4.99	\$0.00	10/2	https://drive.google.com/file/d/1	Yes
	Sweat Bands	1	\$8.39	\$0.00	10/2	https://drive.google.com/file/d/1	Yes
	Build-A-Board ESP32-C3	2	\$0.00	\$20.00	9/30	ElH	Yes
	Battery to Microcontroller connector	1	\$3.59	\$0.00	10/7	https://drive.google.com/file/d/1	Yes
	9V Battery (2 pack)	1	\$8.99	\$0.00	10/7	https://drive.google.com/file/d/1	Yes
	5V step down	1	\$9.99	\$0.00	10/7	https://drive.google.com/file/d/1	Yes

Budget - Part 3

	Rocker Switch	1	\$7.58	\$0.00	10/7	https://drive.google.com/file/d/1	Yes
Shipping	GoBuilda Order 9/30		\$6.99	\$0.00			
	GoBuilda Order 10/2		\$0.00	\$0.00			
	Amazon 10/2 and 10/7		\$0.00	\$0.00			
	McMaster Carr 10/2		\$9.21	\$0.00			
	Amazon 7 day prime trail		\$1.99	\$0.00			
Total			692.62	143.52			

Total Spent:
 Amount Remaining On Card: \$379.80
 Amount Remaining in Budget: **\$287.75**

Appendix: Budget Performance

Starting Amount		\$1,000.00
Amount Remaining on Card		\$307.38
Amount Remaining in Budget		\$163.68
Subsystem Breakdown	Allocated Cost	Actual Cost
Drivetrain	\$326	\$483.71
Chassis	\$150	\$152.99
Pill Dispenser	\$75	\$80.62
Koozie	\$60	\$11.64
Wearable Devices	\$150	\$88.99
Shipping	\$20	\$18.19
Total Spent		\$836.14

Appendix

Customer Requirements

1. Completed final system in a timely manner.
2. Cost effective.
3. Made from durable materials.
4. Medication delivery system that comprises of three individual devices to deliver medications to the correct patients in hospitals.
5. Set anatomical dimensions for Patient A and B.
6. Delivery device must be able to travel from the pill dispensary location to the patient.
7. Biomechanical interaction with an upper extremity of Patient A to signal the delivery system.
8. Thermal interaction with Patient B to signal the delivery system at high body temperatures.
9. The system needs to be maneuverable and agile to navigate the course when delivering medication.
10. Medication delivery should move and deposit a set number of pills to the correct patient zone.
11. Efficient in terms of both time and distance traveled.
12. Visually appealing.
13. Easy for medical staff to handle and store.

Appendix

Engineering Specifications

1. All systems must be fully functional by December 6th, 2025
2. Collective system must remain under a \$1,000 budget
3. The collective system must contain 0 components made of wood
4. The collective system must use 0 adhesion materials (duct tape, glue, etc.)
5. Mechanical fasteners (screws, nuts, bolts, etc.) must be used to secure 100% of the components
6. The chassis frame must be able to support up to a 25 lb load
7. Delivery device must not exceed 1ft x 1ft base area
8. Wearable device must fit subject A's dimensions:
 - a. Thumb circumference- 2.75 inches, pinky circumference- 1.8 inches, wrist circumference- 7 inches, length between top of thumb and wrist- 5 inches, length between top of pinky and thumb- 5.5 inches
Measured the Patient's Dimensions- David Shunk
9. Wearable device must fit subject B's dimensions:
 - a. Thumb circumference- 2.25 inches, pinky circumference- 1.6 inches, wrist circumference- 6 inches, length between top of thumb and wrist- 4.5 inches, length between top of pinky and thumb- 5 inches
Measured the Patient's Dimensions- Leah Kern
10. Delivery device must be able to travel 40 feet in a linear path within a 5% margin of error (2 feet).
<https://equine.ca.uky.edu/content/what-does-%E2%80%98statistically-significant%E2%80%99-actually-mean>
11. Delivery device must be able to turn 90 degrees to left and right with 1% margin of error.

Appendix

Engineering Specifications Continued

12. Delivery device must be able to record the distance traveled while delivering medication using an odometer
 13. Delivery devices must display the total distance traveled while delivering medication
 14. Delivery devices must indicate which patient it is responding to via an LED that operates with 3.3 V and 16 mA
<https://thepihut.com/blogs/raspberry-pi-tutorials/27968772-turning-on-an-led-with-your-raspberry-pis-gpio-pins>
 15. Wearable device must be capable of thermal measurements from 98 to over 102 degrees Fahrenheit in an appropriate/non-invasive way
<https://www.mayoclinic.org/diseases-conditions/fever/in-depth/fever/art-2005099>
 16. Wearable device must be capable of thermal measurements with an accuracy of 1°F or more
[https://www.dwyeromega.com/en-us/resources/thermistor#:~:text=Thermistors%20are%20highly%20accurate%20\(ranging,not%20change%20significantly%20with%20age.](https://www.dwyeromega.com/en-us/resources/thermistor#:~:text=Thermistors%20are%20highly%20accurate%20(ranging,not%20change%20significantly%20with%20age.)
 17. At 102 degrees Fahrenheit, the wearable device must be triggered to send a 5 GHz Wifi signal to the delivery robot
<https://www.mayoclinic.org/diseases-conditions/fever/in-depth/fever/art-2005099>
 18. Wearable device must be capable of detecting a non-functional biomechanical movement of a patient's upper extremity that has 1 degree of freedom through sensors or buttons.
 19. Once biomechanical movement is detected, the wearable device must be capable of sending a 5 GHz Wifi signal to the delivery device
 20. The signal between the delivery device and wearable devices should be able to span over 50ft
 21. Delivery device must be able to travel a total of 200ft
 22. The velocity of the delivery device should be under 5mph
<https://www.sciencedirect.com/science/article/pii/S0921889010000680>
 23. The delivery device must be able to run for 10 minutes. (3 rounds of a 200ft course at an average of 3mph times a factor of safety of 1.5)
- 1.5) [Developing a mobile robot for transport applications in the hospital domain - ScienceDirect](#)
-

Appendix

Engineering Specifications Continued

24. Delivery device must successfully engage with the docking system 95% of the trials
 25. The docking station must interface with a solo cup of the following dimensions 100% of the time
 - a. The cup has a height of 3", outer diameter of 3.5" at the opening, and inner diameter 2.25" at the base
 26. Delivery device must individually dispense 2 correct pills into the proper patient's cup
 27. Delivery device must have the ability to hold at minimum 20 "pills" of size $\frac{5}{8}$ inches in length and $\frac{3}{8}$ inches in diameter
<https://candylandstore.com/shop/jewel-jelly-belly-beans/>
 - a. In terms of volume, the delivery device must be able to hold
 - b. 4.9677 in³ volume for the "pills"
 - i. 20 JB * 1 gal/930 JB * 128 ounces/1 gal * 1.805 in³/1 ounce = 4.9677 in³
<https://www.vcalc.com/wiki/candies-in-container-calculator>
 28. The pill chamber within the delivery device must be able to support 22.6 g which amounts to 20 jelly beans
 - a. 1.13 g/JB * 20 JB = 22.6 g
<https://www.jellybelly.com/general-facts>
 29. The height of the delivery device must not exceed 29", so that it can fit under a hospital bed
<https://www.sondercare.com/learn/hospital-beds/what-is-ideal-height-for-home-hospital-bed/#:~:text=The%20most%20common%20height%20range,%2Dinches%20to%2029%2Dinches.>
 30. The weight of the delivery device must less than 50 lbs
<https://www.bls.gov/ors/factsheet/strength.htm#:~:text=However%2C%20if%20they%20delivered%20slightly,weighing%20less%20than%20one%20pound.>
 31. The weight of the wearable device must less than 5.1 lbs
https://hf.tc.faa.gov/publications/2005-human-factors-guidance-for-the-use-of-handheld/full_text.pdf
 32. The battery in the delivery device should be able to provide 12V at 10 amps and 5V at 3 amps
-

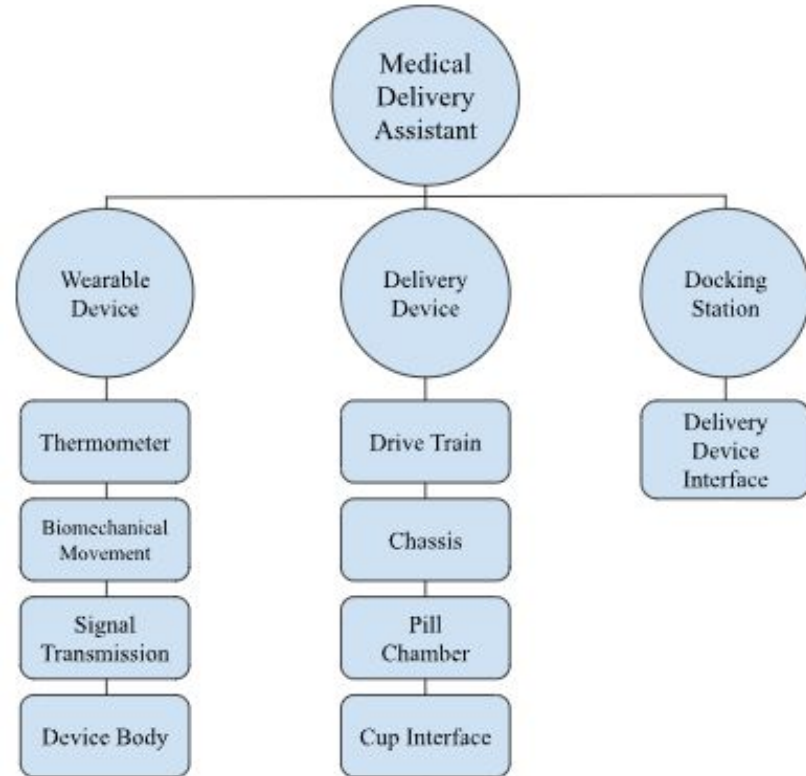
Appendix

Engineering Specifications Continued

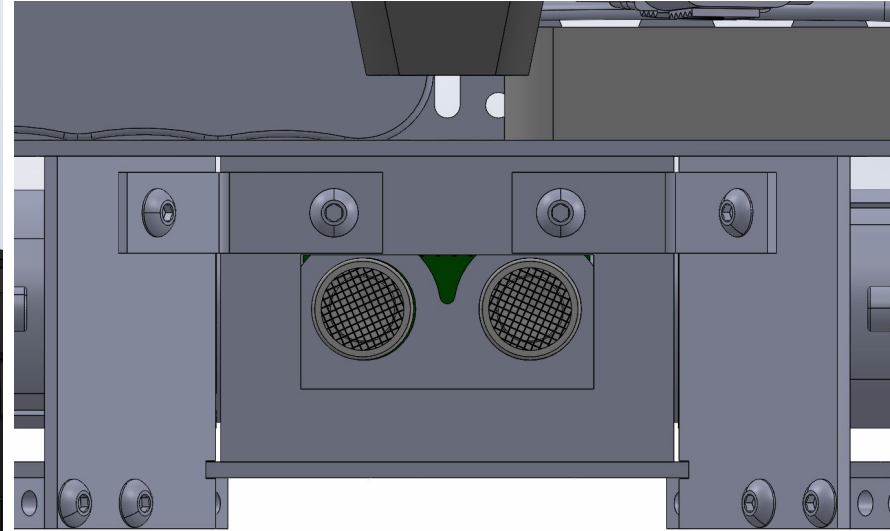
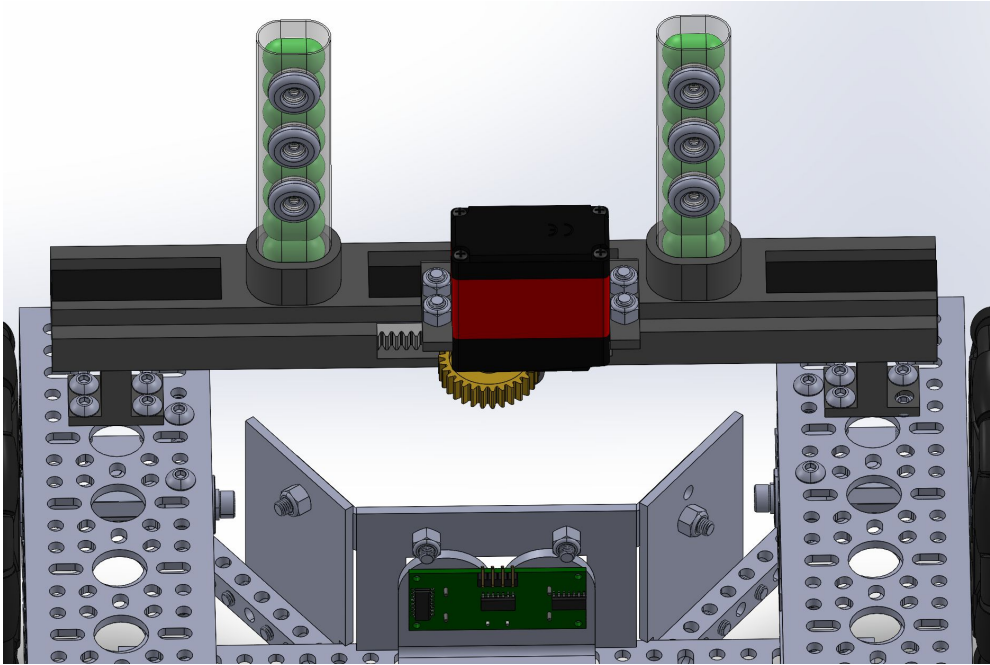
33. The battery in the wearable device should have a maximum voltage of 5V
<https://oxeltech.de/design-guide-for-battery-safety-in-wearable-electronics/>
34. The battery in the wearable device should be able to provide 5V at 3 amps
35. The full system should include 3 microcontrollers, one for the delivery device and one for each wearable device.
36. The battery in the delivery device should be able to be recharged with a charging time of 1 hour
[https://fpvfc.org/beginners-guide-to-lipo-batteries#:~:text=The%20safest%20way%20to%20charge,discharged%20battery%20around%203.2v%20\).](https://fpvfc.org/beginners-guide-to-lipo-batteries#:~:text=The%20safest%20way%20to%20charge,discharged%20battery%20around%203.2v%20).)
37. The microcontrollers must be capable of sending signals via wifi.
38. The microcontrollers must be capable of being powered by a 5V battery supply
39. The full system should be able to complete 9 out of 10 consecutive trials successfully
40. Entire system must be considered aesthetically appealing at a ranking of 3 or higher on the Likert Scale by 3-5 unaffiliated people
41. The wearable device should not cause skin irritation or be uncomfortable via rating 2 or less on a Likert scale of 1-5 by 3-5 unaffiliated people
42. The full system should complete the course in under 3 minutes

Appendix Subsystems

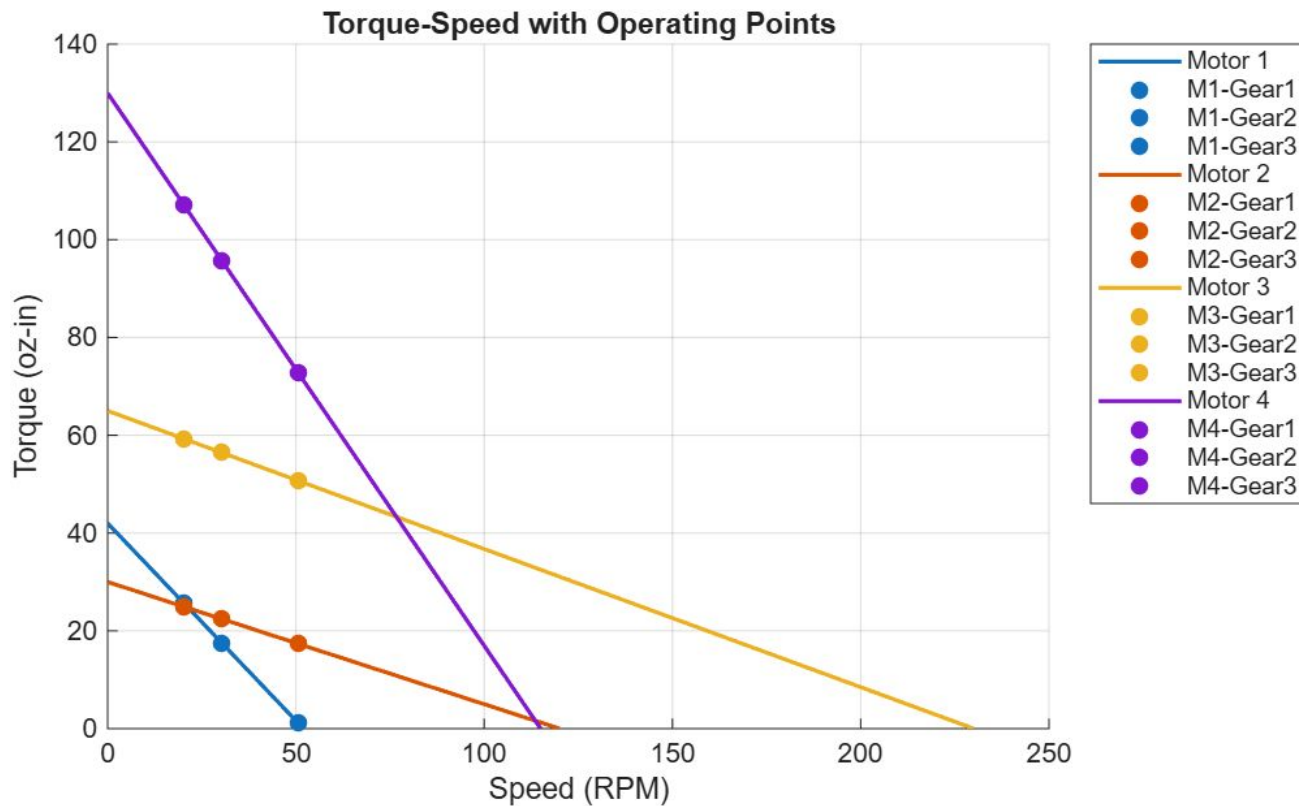
1. Delivery Device
 - a. Drivetrain
 - b. Chassis
 - c. Pill Chamber
 - d. Cup Interface
2. Wearable Device
 - a. Thermometer
 - b. Biomechanical Movement
 - c. Signal Transmission
 - d. Device Body
3. Docking Station
 - a. Delivery Device Interface



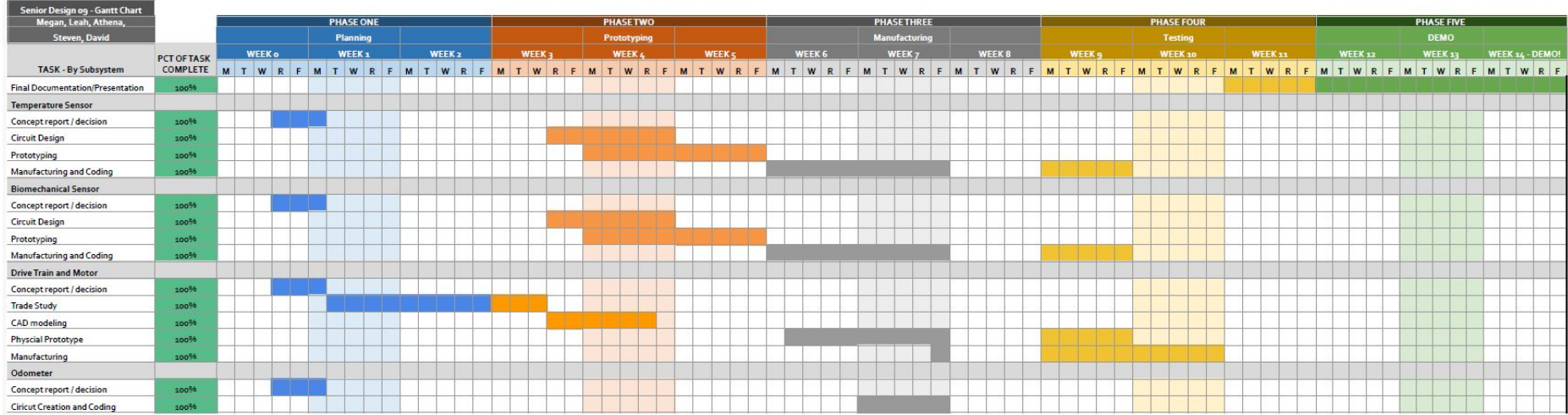
Appendix Completed Virtual Model - Detailed Views



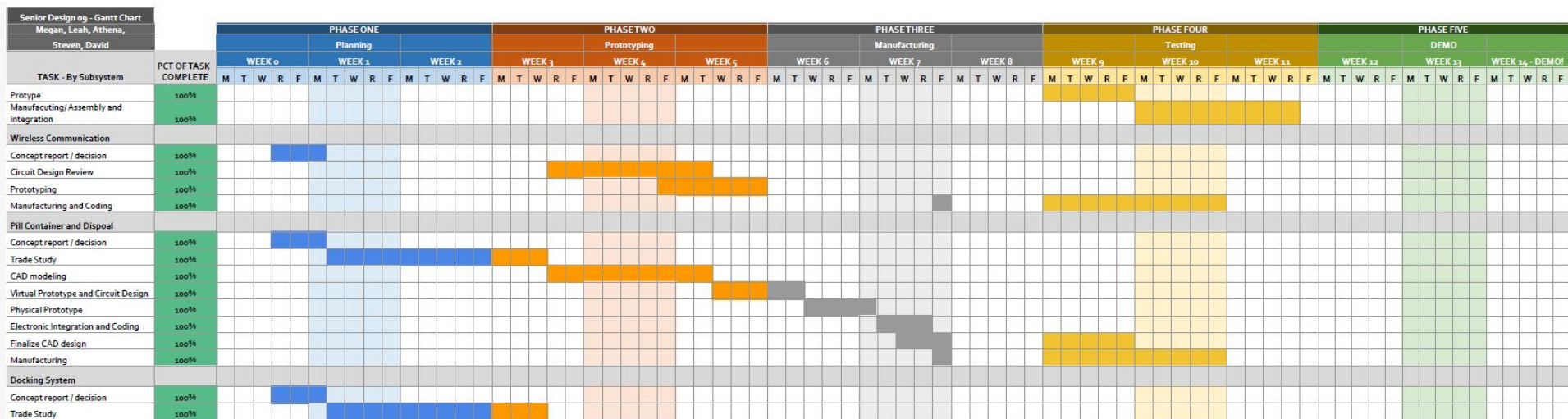
Appendix: Pill Dispenser Torque-Speed Relation



Appendix



Appendix



Appendix

Senior Design 09 - Gantt Chart																																											
Megan, Leah, Athena,		PHASE ONE														PHASE TWO							PHASE THREE							PHASE FOUR							PHASE FIVE						
Steven, David		Planning							Prototyping							Manufacturing							Testing							DEMO													
TASK - By Subsystem		WEEK 0		WEEK 1		WEEK 2		WEEK 3		WEEK 4		WEEK 5		WEEK 6		WEEK 7		WEEK 8		WEEK 9		WEEK 10		WEEK 11		WEEK 12		WEEK 13		WEEK 14 - DEMO!													
PCT OF TASK COMPLETE		M	T	W	R	F	M	T	W	R	F	M	T	W	R	F	M	T	W	R	F	M	T	W	R	F	M	T	W	R	F												
CAD modeling		100%																																									
Prototype		100%																																									
Finalize CAD design		100%																																									
Manufacturing		100%																																									